



Faculty of Technology & Engineering
Devang Patel Institute of Advance Technology and Research
Bachelor of Technology Programme
Computer Science & Engineering

ACADEMIC
REGULATIONS
&
SYLLABUS
(Choice Based Credit System)



Academic Year: 2019-2020



CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Technology and Engineering

ACADEMIC REGULATIONS

Bachelor of Technology (CSE) Programme

(Choice Based Credit System)

Charotar University of Science and Technology (CHARUSAT)

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Year – 2019-2020

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
DEVANG PATEL INSTITUTE OF ADVANCE TECHNOLOGY AND RESEARCH
COMPUTER SCIENCE & ENGINEERING

Vision

To be at the frontiers in the field of computer science and engineering by focusing on areas of research, innovative technology and empowering/ inspiring/ motivating/ encouraging/ educating with the requisite skills to solve the complex technical challenges of modern society and to become influential problem-solvers and critical thinkers to contribute to the well-being of the world.

Mission

The Department of Computer Science and Engineering is committed:

1. To Provide a Conceptual and Functional context in the field of Computer Science and engineering.
2. To inculcate professional ethics, innovative research abilities and leadership qualities.

To build excellent teaching environment for learning and sculpt engineers with deep analytical knowledge base and technical Concepts in the emerging areas of technology.

THE PROGRAMME EDUCATIONAL OBJECTIVES (PEOs)

PEO1	To establish a strong foundation in students and help them to apply engineering fundamentals, mathematics, science and humanities necessary to formulate, analyze, design and implement engineering solutions.
PEO2	To engage in research and find the appropriate solutions for the benefits of mankind through continuous learning, relearning and unlearning.
PEO3	To bring out world class engineers, scientists, entrepreneurs, model citizens, problem solvers and critical thinkers with skills developed by practices of advance technologies.
PEO4	To make students aware about their capabilities so that they can analyze real life problems to develop economically viable and socially acceptable solutions that are sustainable using their technical and analytical skills.
PEO5	To provide an environment where the students are nurtured in every aspect for their holistic development such that they achieve excellence in professionalism, interpersonal skills as well as moral and ethical conduct.
PEO6	To infuse in students, the ability to work as a part of team on multidisciplinary projects and develop leadership qualities that meet the challenging and changing global needs.
PEO7	To prepare and make students ready for presenting themselves on global platform by encouraging them to adapt rapidly changing industry requirements.

PROGRAM OUTCOMES (POs)

At the end of the program, the student will be able to:

PO1	Engineering knowledge: Apply knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet specified needs with appropriate consideration for public health and safety, and the cultural, societal, and environmental considerations.
PO4	Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO7	Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9	Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

At the end of the program, the student will be able to:

PSO1	To instill analytical, technical and ethical skills required to identify optimized IT solutions for complex problems by applying critical and computational thinking.
PSO2	To undergo appropriate research and use innovative technologies for designing IT based solutions that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental benefits.



FACULTY OF TECHNOLOGY AND ENGINEERING
ACADEMIC REGULATIONS
Bachelor of Technology Programmes
Choice Based Credit System

To ensure uniform system of education, duration of undergraduate and post graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

1. *System of Education*

Choice based Credit System with Semester pattern of education shall be followed across The Charotar University of Science and Technology (CHARUSAT) both at Undergraduate and Master's levels. Each semester will be at least 90 working day duration. Every enrolled student will be required to take a course works in the chosen subject of specialization and also complete a project/dissertation if any. Apart from the Programme Core courses, provision for choosing University level electives and Programme/Institutional level electives are available under the Choice based credit system.

2. *Duration of Programme*

(i)	Undergraduate programme	(B.Tech)
	Minimum	8 semesters (4 academic years)
	Maximum	16 semesters (8 academic years)

3. *Eligibility for admissions*

As enacted by Govt. of Gujarat from time to time.

4. *Mode of admissions*

As enacted by Govt. of Gujarat from time to time.

5. *Programme structure and Credits*

As per annexure – 1 attached

6. Attendance

- 61 All activities prescribed under these regulations and listed by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student from attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Dean/Principal.
- 62 Student attendance in a course should be 80%.

7 Course Evaluation

- 71 The performance of every student in each course will be evaluated as follows:
- 71.1 Internal evaluation by the course faculty member(s) based on continuous assessment, for 30% of the marks for the course; and
- 71.2 Final examination by the University through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these, for 70% of the marks for the course.

72 Internal Evaluation

As per Annexure – 1 attached

73 University Examination

- 731 The final examination by the University for 70% of the evaluation for the course will be through written paper and 100% for practical test or oral test or presentation by the student or a combination of any two or more of these.
- 732 **In order to earn the credit in a course a student has to obtain grade other than FF.**

7.4 Performance at Internal & University Examination

- 741 Minimum performance with respect to internal marks as well as university examination will be an important consideration for passing a course. Details of minimum percentage of marks to be obtained in the examinations (internal/external) are as follows

Minimum marks in University Exam per subject	Minimum marks Overall per subject
40%	45%

- 742 A student failing to score 45% of the final examination will get a FF grade.
- 743 If a candidate obtains minimum required marks per subject but fails to obtain minimum required overall marks, he/she has to repeat the university examination till the minimum required overall marks are obtained.

8 Grading

- 8.1 The total of the internal evaluation marks and final University examination marks in each course will be converted to a letter grade on a ten-point scale as per the following scheme:

Table: Grading Scheme (UG)

Range of Marks (%)	≥80	≥73 <80	≥66 <73	≥60 <66	≥55 <60	≥50 <55	≥45 <50	<45
Letter Grade	AA	AB	BB	BC	CC	CD	DD	FF
Grade Point	10	9	8	7	6	5	4	0

- 8.2 The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are calculated as follows:

- (i) $SGPA = \sum C_i G_i / \sum C_i$ where C_i is the number of credits of course i
 G_i is the Grade Point for the course i
and $i = 1$ to n , n = number of courses in the semester
- (ii) $CGPA = \sum C_i G_i / \sum C_i$ where C_i is the number of credits of course i
 G_i is the Grade Point for the course i
and $i = 1$ to n , n = number of courses of all semesters up to which CGPA is computed.
- (iii) No student will be allowed to move further if CGPA is less than 3 at the end of every academic year.
- (iv) A student will not be allowed to move to third year if he/she has not cleared all the courses of first year.
- (v) A student will not be allowed to move to fourth year if he/she has not cleared all the courses of second year.

9 Awards of Degree

- 9.1 Every student of the programme who fulfils the following criteria will be eligible for the award of the degree:
- 9.1.1 He should have earned at least minimum required credits as prescribed in course structure; and
- 9.1.2 He should have cleared all internal and external evaluation components in every course; and
- 9.1.3 He should have secured a minimum CGPA of 4.5 at the end of the programme;
- 9.1.4 In addition to above, the student has to complete the required formalities as per the regulatory bodies, if any.

- 9.2 The student who fails to satisfy minimum requirement of CGPA at the end of program will be allowed to improve the grades so as to secure a minimum CGPA for award of degree. Only latest grade will be considered.

10. Award of Class

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Distinction:	$\text{CGPA} \geq 7.5 \text{ \& } \leq 10.0$
First class:	$\text{CGPA} \geq 6.0 \text{ \& } < 7.5$
Second Class:	$\text{CGPA} \geq 5.0 \text{ \& } < 6.0$
Pass Class:	$\text{CGPA} < 5.0$

11. Transcript

The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

**CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
(CHARUSAT)**

FACULTY OF TECHNOLOGY & ENGINEERING (FTE)

CHOICE BASED CREDIT SYSTEM

FOR

BACHELOR OF TECHNOLOGY & ENGINEERING

A. Choice Based Credit System:

With the aim of incorporating the various guidelines initiated by the University Grants Commission (UGC) to bring equality, efficiency and excellence in the Higher Education System, Choice Based Credit System (CBCS) has been adopted. CBCS offers wide range of choices to students in all semesters to choose the courses based on their aptitude and career objectives. It accelerates the teaching-learning process and provides flexibility to students to opt for the courses of their choice and / or undergo additional courses to strengthen their Knowledge, Skills and Attitude.

1. CBCS – Conceptual Definitions / Key Terms (Terminologies)

1.1. Core Courses

1.1.1 University Core (UC)

University Core Courses are those courses which all students of the University of a Particular Level (PG/UG) will study irrespective of their Programme/specialisation.

1.1.2 Programme Core (PC)

A ‘Core Course’ is a course which acts as a fundamental or conceptual base for Chosen Specialisation of Engineering. It is mandatory for all students of a particular Programme and will not have any other choice for the same.

1.2 Elective Course (EC)

An ‘Elective Course’ is a course in which options / choices for course will be offered. It can either be for a Functional Course / Area or Streams of Specialization / Concentration which is / are offered or decided or declared by the University/Institute/Department (as the case may be) from time to time.

1.2.1 Institute Elective Course (IE)

Institute Courses are those courses which any students of the University/Institute of a Particular Level (PG/UG) will choose as offered or decided by the University/Institute from time-to-time irrespective of their Programme /Specialisation

1.2.2 Programme Elective Course (PE):

A ‘Programme Elective Course’ is a course for the specific programme in which students will opt for specific course(s) from the given set of functional course/ Area or Streams of Specialization options as offered or decided by the department from time-to-time

1.2.3 Cluster Elective Course (CE):

An ‘Elective Course’ is a course which students can choose from the given set of functional course/ Area or Streams of Specialization options (eg. Common Courses to EC/CE/IT/EE) as offered or decided by the Institute from time-to-time.

1.3 Non Credit Course (NC) - AUDIT Course

A 'Non Credit Course' is a course where students will receive Participation or Course Completion certificate. This will not be reflected in Student's Grade Sheet. Attendance and Course Assessment is compulsory for Non Credit Courses

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

TEACHING & EXAMINATION SCHEME FOR B TECH PROGRAMME IN CE ENGINEERING

CHOICE BASED CREDIT SYSTEM

Sem	Course Code	Course Title	Teaching Scheme				Examination Scheme				
			Contact Hours			Credit	Theory		Practical		Total
			Theory	Practical	Total		Internal	External	Internal	External	
Sem 3	MA245	Discrete Mathematics	4	0	4	4	30	70	0	0	100
	CE251	Java Programming	3	4	7	5	30	70	50	50	200
	CE252	Digital Electronics	3	2	5	4	30	70	25	25	150
	CE253	Data Communication & Networking	3	2	5	4	30	70	25	25	150
	XXXXX	University Elective - I	2			2	30	70			100
	CE244	Software Group Project-I	0	4	4	2			50	50	100
	HS122 A	Values and Ethics	2	0	2	2		100			100
			17	12	29	23	120	280	180	220	900
Sem 4	CE245	Data Structures & Algorithms	3	2	5	4	30	70	25	25	150
	CE246	Database Management System	4	4	8	6	30	70	50	50	200
	CE247	Web Technologies	2	4	6	4	30	70	50	50	200
	CE248	Operating System	3	2	5	4	30	70	25	25	150
	CE256	Computer Organization	3	0	3	3	30	70	0	0	100
	XXXXX	University Elective - II	2			2	30	70			100
	CE255	Software Group Project-II	0	4	4	2			50	50	100
	HS133 A	CREATIVITY, PROBLEM SOLVING AND INNOVATION	2	0	2	2		100			100
			19	16	35	27	150	350	230	270	1100

List of University Electives			
Code	University Elective - I (UE - I)	Code	University Elective - II (UE - II)
EC281.01	Introduction to MATLAB Programming	EC282.01	Prototyping Electronics with Arduino
CE281.01	Art of Programming	CE282.01	Web Designing
CL281.01	Environmental Sustainability and Climate Change	CL282.01	Basics of Environmental Impact Assessment
EE286	Python for Electrical Engineering	EE283	Computer Programming for Electrical Engineering
IT281.01	ICT Resources and Multimedia	IT282.01	Internet Technology and Web Design
ME281.01	Engineering Drawing	ME282.01	Material Science
PH233	Fundamentals of Packaging	PH238	Cosmetics in daily life
PD260	Basic Laboratory Techniques	NR261	Life Style Diseases & Management
NR251	First Aid & Life Support	PT192	Occupational Health & Ergonomics
PT191	Health Promotion and Fitness	CA225	Programming the Internet
CA224	Introduction to Web Designing	BM241	Health Care Management
BM231	Banking and Insurance		

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
TEACHING & EXAMINATION SCHEME FOR B TECH PROGRAMME IN CSE
CHOICE BASED CREDIT SYSTEM

Sem	Course Code	Course Title	Teaching Scheme					Examination Scheme				
			Contact Hours				Credit	Theory		Practical		Total
			Theory	Practical	Tutorial	Total		Internal	External	Internal	External	
Third Year Sem 5	CS341	Artificial Intelligence	3	2	0	5	4	30	70	25	25	150
	CE341	Microprocessor Architectures & Assembly Programming	3	2	0	5	4	30	70	25	25	150
	CE342	Design & Analysis of Algorithms	3	2	0	5	4	30	70	25	25	150
	CE344	Computer Networks	3	2	0	5	4	30	70	25	25	150
	CSXXX	Programme Elective-I	2	4	0	6	4	30	70	50	50	200
	CS348	Software Group Project-III	0	2	0	2	2	-	-	50	50	100
	CS343	Summer Internship-I			0	0	3	0	0	75	75	150
	HS124.01A	Professional Communication	0	2	0	2	2	0	0	30	70	100
		14	16	0	30	27	150	350	305	345	115	
Third Year Sem 6	CS344	Machine Learning	3	2	0	5	4	30	70	25	25	150
	CS345	Cryptography & Network Security	4	2	0	6	5	30	70	25	25	150
	CE349	Theory of Computation	3	0	0	3	3	30	70	0	0	100
	CS346	Software Engineering	3	2	0	5	4	30	70	25	25	150
	CSXXX	Programme Elective-II	3	2	0	4	4	30	70	25	25	150
	CS349	Software Group Project-IV	0	4	0	4	4	-	-	100	100	200
	HS134 A	Contributor Personality Development	0	2	0	2	2	30	70	0	0	100
			16	14	0	30	26	180	420	200	200	100

Code	Programme Elective - I (PE - I)
CS376	Advanced Java Programming
CS371	Digital Image Processing
CS372	Advanced Web Technology

Code	Programme Elective - II (PE - II)
CS373	Internet of Things
CS375	Service Oriented Computing
CS374	Modern Networks

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY											
TEACHING & EXAMINATION SCHEME FOR B TECH PROGRAMME IN CSE											
CHOICE BASED CREDIT SYSTEM											
Sem	Course Code	Course Title	Teaching Scheme				Examination Scheme				
			Contact Hours			Credit	Theory		Practical		Total
			Theory	Practical	Total		Internal	External	Internal	External	
Final Year Sem 7	CS441	Programming in Python	3	2	5	4	30	70	25	25	150
	CS442	Data Science and Analytics	3	4	7	5	30	70	50	50	200
	CE442	Design of Language Processors	4	2	6	5	30	70	25	25	150
	CE443	Cloud Computing	3	2	5	4	30	70	25	25	150
	CSXXX	Programme Elective-III	4	2	6	5	30	70	25	25	150
	CS445	Software Group Project-IV	0	2	2	1	0	0	25	25	50
	CS446	Summer Internship - II			0	3	0	0	75	75	150
			17	14	31	27	150	350	250	250	1000
Final Year Sem 8	CS447	Software Project Major	0	36	36	20			250	350	600
			0	36	36	20			250	350	600

LIST OF ELECTIVE SUBJECTS FOR FOURTH YEAR B TECH PROGRAMME IN COMPUTER Science & ENGINEERING		
ELECTIVES	Code	Elective - III
	CS471	Mobile Computing
	CS472	Cyber Security and Cyber Laws
	CS473	Blockchain Technology & Applications

Note:

- **University Elective (UE):** - University Electives are offered in common slots and offered by various departments. Students of any programme can select these electives. Subjects like Research Methodology, Occupational Health & Safety, Engineering Economics, Professional Ethics, and Project Management, Disaster Management, Risk Management etc. can be included.
- **Cluster Elective (CT):** - Institutional Electives means common electives among a cluster of programmes (eg. CE/IT/EC/EE etc.). If Institutional Electives are not applicable, it will be Programme electives
- Programme Elective (PE): -
- **Institute Elective (IE):**-
- Provision for Auditing a course will be available
- Audit courses may be offered and decided based on need of the institute/programs.

B. Tech. (Computer Science & Engineering) Programme

SYLLABI (Semester – 3)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

MA245: DISCRETE MATHEMATICS

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	4	0	-	4	4
Marks	100	0	-	100	

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Set Theory and Predicate calculus	08
2.	Relations, Lattices and Boolean algebra	14
3.	Abstract Algebra	10
4.	Graphs and Graph Algorithms	12
5.	Matrix Algebra- II	10
6.	Fundamentals of Finite State Machine and Recurrence	06

Total hours (Theory): 60

Total hours (Lab): 00

Total hours: 60

B. Detailed Syllabus:

- | | |
|--|---------------------|
| 1 Set Theory and Predicate Calculus | 08 Hours 14% |
| 1.1. Proposition, Types of Proposition, Tautology, Contradictions, Connectives, Types of connectives | |
| 1.2. Logical equivalence, Verification using truth table | |
| 1.3. Converse, Inverse and Contrapositive, Normal forms | |
| 1.4. Introduction to predicates and quantifiers | |
| 1.5. Predicate calculus using rules of inferences | |
| 1.6. Logic in proof, Methods of Proof, | |
| 1.7. Mathematical Inductions (First Principle) | |
| 1.8. Properties of set operations with Predicate logic. | |
| 1.9. Cardinality of sets, Cartesian product of sets | |
| 2 Relations, Lattice and Boolean Algebra | 14 Hours 22% |
| 2.1. Relations on sets, Types of Relations in a set | |
| 2.2. Properties of Relations | |
| 2.3. Representations of Relations | |
| 2.4. Equivalence relation, Covering of a set, Partition of a set | |
| 2.5. Partially ordered relations, Partially ordered sets, Lattice, Sub lattices | |
| 2.6. Properties of lattice | |
| 2.7. Some Special Lattices | |
| 2.8. Finite Boolean Algebra, atoms, anti - Atoms, | |
| 2.9. Sub - Boolean algebra, Boolean Expression | |
| 2.10. Boolean Functions, | |
| 2.11. Canonical Forms, Karnaugh map representation, Quine Mckausky's Algorithm | |
| 3 Abstract Algebra | 10 Hours 16% |
| 3.1. Groupoid, Semi group, Monoid, Group | |
| 3.2. Order of group, order of an element, Lagrange's theorem | |
| 3.3. Subgroup, Cyclic subgroup, Permutation Group | |
| 3.4. Introduction to Ring Theory | |
| 3.5. Sub ring, Ring Homomorphism, | |
| 3.6. Ideals | |

4 Graphs and Graph Algorithms**12 Hours 20%**

- 4.1. Basic terminologies, Simple graph, Types of graphs
- 4.2. Degree of a vertex
- 4.3. Sub graphs, Spanning Sub-graphs, Isomorphic graphs
- 4.4. Path and connectivity
- 4.5. Eulerian and Hamiltonian graph
- 4.6. Matrix Representation of graph
- 4.7. Planar Graphs
- 4.8. Introduction to tree, Directed tree, Forest
- 4.9. Types of trees, Spanning Tree, Minimal Spanning Tree
- 4.10. Algorithm to find minimal spanning tree, Prim's, Krushkal's and Dijkstra's Algorithm

5 Matrix Algebra -II**10 Hours 16%**

- 5.1. Revision of Determinant and Matrix
- 5.2. Eigen values and Eigen vectors of Matrices
- 5.3. Eigen values and Eigen vectors of Special Matrices
- 5.4. Applications of Cayley - Hamilton Theorem

6 Basics of Finite State Machine and Recurrence Relation**06 Hours 12%**

- 6.1. Introduction to Strings, Languages,
- 6.2. Regular expression, Grammars
- 6.3. Introduction to Recurrence relation,
- 6.4. Generating function

- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval. It carries a weightage of 5%.
- Two Quizzes (surprise tests) will be conducted which carries 5% component of the overall evaluation.
-

Course Outcome (COs):

- At the end of the course, the students will be able to

CO1	Develop logical argument using truth table and rules of inferences in predicate calculus.
CO2	Relation and types of relations define on sets and utilize it to construct Hasse diagram and lattices on sets.

CO3	Different algebraic structures like groupoid, semi group, monoid, group , cyclic group and permutation group
CO4	Graph and types of the graphs and identify the real world phenomena in terms of graph theory.
CO5	find and explain significant of Eigenvalues and Eigenvectors of a square matrix, use Cayley-Hamilton's theorem to find inverse and power of a square matrix.
CO6	The concept of recurrence, generating functions and their applications in solving recurrence relations.

• **Course Articulation Matrix:**

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO 1	PSO 2
CO1	3	3	1	-	1	-	-	-	-	-	-	-	2	-
CO2	3	2	-	-	-	-	-	-	-	-	-	-	3	-
CO3	3	1	-	-	-	-	-	-	-	-	-	-	2	-
CO4	3	2	1	-	2	-	-	-	-	-	-	-	3	1
CO5	3	1	-	-	-	-	-	-	-	-	-	-	3	-
CO6	3	-	-	-	1	-	-	-	-	-	-	-	3	-

- Correlation levels 1, 2 or 3 as defined below:
- 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

D. Recommended Study Material:

❖ **Reference Books:**

1. Rosen, Kenneth H., and Kamala Krithivasan. Discrete mathematics and its applications. Vol. 6. New York: McGraw-Hill, 1995.
2. Tremblay, Jean-Paul, and Rampurkar Manohar. Discrete mathematical structures with applications to computer science. New York: McGraw-Hill, 1975.
3. McAllister, D. F., and D. F. Stanat. "Discrete Mathematics in Computer Science." 1977 Prentice-Hall, Inc
4. Deo, Narsingh. Graph theory with applications to engineering and computer science. Courier Dover Publications, 2016.
5. B. Kolman and R. C. Busby, Discrete Mathematical Structures for Computer Science, 2nd edition, Prentice-Hall, Englewood Cliffs, New Jersey (1987).

6. Malik, D. S., and Mridul K. Sen. Discrete mathematical structures: theory and applications. Course Technology, 2004.
7. Thomas H. Cormen, Leiserson, C. E., Rivest, R. L., & Stein, C. Introduction to algorithms (Vol. 6). Cambridge: MIT press, 2001
8. Anton, Howard. Elementary linear algebra. John Wiley & Sons, 2010.
9. Gallian, Joseph. Contemporary abstract algebra. Cengage Learning, 2016

❖ **Web Materials:**

1. <http://mathworld.wolfram.com/>
2. <http://en.wikipedia.org/wiki/Math>

CE251: JAVA PROGRAMMING

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	4	-	7	5
Marks	100	100	-	200	

Pre-requisite courses:

- Basic programming skill.

A. Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Fundamental of Programming in Java	02
2.	Class Fundamentals	04
3.	Array & String Handling	02
4.	Inheritance, Interfaces & Packages	06
5.	Exceptions Handling	05
6.	Multithreaded Programming	07
7.	File I/O and NIO	07
8.	Java Collection Frameworks and Generics	12
	Total hours (Theory) :	45
	Total hours (Lab) :	60
	Total hours :	120

B. Detailed Syllabus:

1.	Fundamental of Programming in Java	02 Hours	04%
	History of Java, Basic overview of java, Bytecode, JVM Buzz-words, Application and applets, Constants, Variables & Data Types, Comment, Operators, Control Flow		
2.	Class Fundamentals	04 Hours	09%
	General form of class, Creating class Overloading methods, Constructor ,Declaring Object, Returning objects, using objects as parameters, Assigning object reference		

	variables, Introducing Access control , Understanding static Introducing final, The finalize () method,The this keyword ,Garbage collection		
3.	Array & String Handling	02 Hours	04%
	Array basics, String Array, String class, StringBuffer and StringBuilder class, String Tokenizer Class and Object Class		
4.	Inheritance, Interfaces & Packages	06 Hours	14%
	Inheritance: Using super creating multilevel Hierarchy, method overriding, Dynamic method dispatch, abstract classes, Using final with Inheritance, Using Package: Defining package, Finding package and CLASSPATH, Access protection, Importing package, Interface: Defining Interface, Default Methods, Implementing Interface, Variables in Interface		
5.	Exceptions Handling	05 Hours	11%
	Exception types, Try ...Catch...Finally, Throw, Throws, creating your own exception subclasses		
6.	Multithreaded Programming	07 Hours	16%
	Life cycle of thread, thread methods, thread priority, thread exceptions, Implementing Runnable interface, Synchronization and Concurrency		
7.	File NIO	07 Hours	16%
	File and Directories, Byte streams and character streams, Random Access Files,NIO: Meta Data File Attributes Buffers, Channels, Recursive Operation.		
8.	Collection Framework and Generics	12 Hours	26%
	Collections of objects, Collections: Sets, Sequence, Map, Understanding Hashing, Use of Array List & Vector, Generics Class, Lamda Expression, Functional Reference, Method Reference, Optional Classes, Processing data with streams		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand and implement Object Oriented programming concept using basic syntaxes of control Structures, strings and function for developing skills of logic building activity and garbage collection for saving resources.
CO2	Demonstrate basic problem solving skills: analyzing problems, modeling a problem as a system of objects, creating algorithms, and implementing models and algorithms in an object-oriented computer language (classes, objects, methods with parameters, abstract classes, interfaces, inheritance and polymorphism).
CO3	Use and develop Concurrency theory: progress guarantees, deadlock, livelock, starvation, linearizability.
CO4	Build and test program using new IO api and exception handling
CO5	Analyze and apply collection framework and generics to solve different data structure algorithms.
CO6	Understand and apply java new features

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	2	3	3	2	-	-	-	-	1	-	-	-	2	-
CO3	2	3	3	3	-	-	-	-	-	-	-	-	2	-
CO4	1	2	3	2	-	-	-	1	1	-	-	-	2	-
CO5	1	2	2	2	-	-	-	-	-	-	-	-	2	-
CO6	1	2	-	-	-	-	-	-	-	-	-	1	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

C. Recommended Study Material:**❖ Text Books:**

1. Java: The Complete Reference, Ninth Edition by Herbert Schildt , Oracle Press.
2. Java 8 in Action: Lambdas, Streams, and Functional-style Programming by Alan Mycroft and Mario Fusco, Manning Publication

❖ Reference Books:

1. Thinking in Java, Bruce Eckel, Prentice Hall

2. Java: A Beginner's Guide (Sixth Edition) by Herbert Schildt , Oracle Press.
3. Core Java Volume I--Fundamentals (9th Edition) (Core Series) by Cay S. Horstmann, Prentice Hall

❖ **Web Materials:**

1. <https://docs.oracle.com>

CE252: DIGITAL ELECTRONICS

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	4
Marks	100	50	-	150	

Pre-requisite courses:

- Basic Electronics

A. Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Number Systems	5
2.	Boolean Algebra and Logic Gates	5
3.	Simplification of Boolean Functions	6
4.	Combinational Logic	5
5.	Combinational Logic With MSI AND LSI	5
6.	Sequential Logic	10
7.	Registers, Counters and the Memory Unit	5
8.	Processor Logic Design	4
	Total hours (Theory) :	45
	Total hours (Lab) :	30
	Total hours :	75

B. Detailed Syllabus:

1.	Number Systems	05 Hours	10%
	Digital computer and digital systems, Binary Number, Number base conversion Octal and Hexadecimal Number, Complements, Binary Codes, Binary Storage and register, Binary Logic, Integrated Circuit		
2.	Boolean Algebra and Logic Gates	05 Hours	10%
	Basic Definition, Axiomatic Definition of Boolean Algebra, Minterm And Maxterms, Basic Theorem and Properties of Boolean Algebra, Logic Operations, Digital Logic Gates, IC digital Logic Families		
3.	Simplification of Boolean Functions	06 Hours	12%
	Two-Three Variable K-map, Four- Five Variable K-map, Product of sum Simplification, NAND or NOR implementation, Don't Care condition 3.4 Tabulation method		
4.	Combinational Logic	05 Hours	12%

	Introduction, Design Procedure, Hazards, Adder, subtractor, Code Conversion, Universal Gate, exclusive OR & equivalence functions		
5.	Combinational Logic With MSI and LSI	05 Hours	12%
	Introduction, Binary Parallel Adder, Decimal Adder, Magnitude Comparator, Decoder, Multiplexer, ROM, PLA, PAL		
6.	Sequential Logic	10 Hours	22%
	Introduction, RS,JK,D,T Flip-Flops, Triggering of Flip-Flops, Flip-Flop Excitation Tables, Analysis of Clocked Sequential Circuits, State Reduction and Assignment Design Procedure, Design of Counters, Design with State Equations		
7.	Registers, Counters and the Memory unit	05 Hours	12%
	Introduction, Registers, Shift Registers, Ripple Counters, Synchronous Counters, Timing Sequences, Memory Unit, Johnson counter		
8.	Processor Logic Design	04 Hours	10%
	Processor Organization, Arithmetic Logic Unit, Design of ALU, Status Register, Design of Shifter, Processor Unit		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand, convert and examine the structure of various number systems and its application in digital design.
CO2	Simplification of Boolean function and its conversion in different forms.
CO3	Understand, analyse and design various combinational circuits.
CO4	Compare different flip-flop characteristics and design of Flip-Flop with gates. With FF Design, understand and analyse various sequential circuits.
CO5	Explain the basic requirements of ALU design and its variation.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	1	-	-	-	-	-	-	-	-	-	-	1	-
CO2	3	1	-	-	-	-	-	-	-	-	-	-	1	-
CO3	3	1	-	-	1	-	-	-	-	-	-	-	1	-
CO4	2	2	-	-	1	-	-	-	-	-	-	-	1	-
CO5	2	-	-	-	1	-	-	-	-	-	-	-	1	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

C. Recommended Study Material:

❖ **Text book:**

1. Digital Logic and Computer Design By M Morris Mano, PHI- Publication 2002.
2. Digital Principles and Applications By Malvino & Leach, Seventh Edition, McGraw-Hill Education.

❖ **Reference book:**

1. Digital Design M. Morris Mano and Michael D. Ciletti, Pearson Education.

CE253: DATA COMMUNICATION & NETWORKING

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	4
Marks	100	50	-	150	

Pre-requisite courses:

- N/A

A. Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction and Basic Concepts	02
2.	The OSI Model	02
3.	Signals	04
4.	Encoding And Modulation	06
5.	Data Transmission	07
6.	Multiplexing	08
7.	Error Correction and Detection	05
8.	Data Link Control	05
9.	Local Area Networks	03
10.	Switching	03
	Total hours (Theory) :	45
	Total hours (Lab) :	30
	Total hours :	75

B. Detailed Syllabus:

1.	Introduction and Basic Concepts	02 Hours	04%
	Why to study data communication?, Data Communication, Networks, Protocols and Standards, Standards Organizations		
2.	The OSI Model	02 Hours	04%
	The Model, Functions of the layers, TCP/IP Protocol Suites		
3.	Signals	04 Hours	11%

	Analog and Digital, Periodic and A periodic Signals, Analog Signals, Time and Frequency Domains, Composite Signals, Digital Signals		
4.	Encoding And Modulation	06 Hours	14%
	Digital to Digital Conversion, Analog to Digital Conversion, Digital to Analog Conversion, Analog to Analog Conversion		
5.	Data Transmission	07 Hours	18%
	Digital data transmission, DTE-DCE Interface, Modems, 56K Modems, Cable Modems, Guided Media, Unguided Media, Transmission Impairments, Performance Wavelength, Shannon Capacity, Media Comparison, PSTN		
6.	Multiplexing	08 Hours	14%
	Many to one/one to Many, Frequency division Multiplexing, Wave division Multiplexing, Time division Multiplexing, Multiplexing applications		
7.	Error Correction and Detection	05 Hours	09%
	Types of Errors, Detection, Parity Check, Vertical Redundancy Check, Longitudinal Redundancy Check, Cyclic Redundancy Check, Checksum, Error Correction		
8.	Data Link Control	05 Hours	09%
	Line Discipline, Flow Control, Error Control		
9.	Local Area Networks	03 Hours	04%
	802, Ethernet, Other Ethernet Networks, Token Bus, Token Ring, FDDI		
10.	Switching	03 Hours	07%
	Circuit Switching, Packet Switching		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Illustrate concept, architecture and organisation of computer networks
CO2	Analyse existing networking standards and implementations
CO3	Evaluate key networking protocols, and their hierarchical relationship in the

	context of a conceptual model, such as the OSI and TCP/IP framework
CO4	Understand how signals are used to transfer data between nodes
CO5	Identify the different types of network devices and their functions within a network
CO6	Create multiple host and network architectures, given business requirements and constraints

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	1	-	1	-	-	-	-	-	-	-	2	-	-
CO2	1	2	1	1	2	-	-	-	-	-	-	2	1	1
CO3	3	2	2	1	2	-	-	-	-	1	-	2	1	1
CO4	2	3	1	1	1	-	-	-	-	2	-	2	-	-
CO5	3	2	3	1	3	-	-	-	1	1	-	3	1	1
CO6	3	3	3	2	3	-	-	-	2	2	-	3	3	2

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

C. Recommended Study Material:

❖ Text book:

1. Data communication & Networking, Bahrouz Forouzan, McGraw-Hill

❖ Reference book:

1. Data and Computer Communications, William Stallings, Prentice Hall
2. Computer Network, Andrew S. Tanenbaum, Fourth Edition, Prentice Hall

❖ Web material:

1. www.wikipedia.org
2. <http://www.webopedia.com>

❖ Software:

1. Wireshark
2. Cisco Packet Tracer

CE244: SOFTWARE GROUP PROJECT-I

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	0	4	-	4	2
Marks	0	100	-	100	

Pre-requisite courses:

- Programming Language, Software Engineering.

A. Outline of the Course:

- Student at the beginning of a semester may be advised by his/her supervisor (s) for recommended courses.
- Students will work together in a team (at most three) with any programming language.
- Students are required to get approval of project definition from the department.
- After approval of project definition students are required to report their project work on weekly basis to the respective internal guide.
- Project will be evaluated at least once per week in laboratory Hours during the semester and final submission will be taken at the end of the semester as a part of continuous evaluation.
- Project work should include whole SDLC of development of software / hardware system as a solution of particular problem by applying principles of Software Engineering.
- Students have to submit project with following listed documents at the time of final submission.
 - a. Final Project Report
 - b. Project Setup file with Source code
 - c. Project Presentation (PPT)
- A student has to produce some useful outcome by conducting experiments or project work.

Total hours (Theory) : 00

Total hours (Lab) : 30

Total hours : 30

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Create enhanced employment by moulding the students with higher technical skill.
CO2	Promote creative thinking, to provide hands on actual technology.
CO3	Handle software project and get use to software development processes.
CO4	Correlate knowledge of different subjects and apply theoretical knowledge to implement project for identified problem.
CO5	Write technical report and deliver presentation by applying different visualization tools and evaluation metrics.
CO6	Improve communication and presentation skill.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	3	-	-	-	-	-	-	-	-	1	-	-	-
CO2	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO3	-	-	-	-	2	-	-	-	2	-	-	-	-	-
CO4	3	-	-	-	-	-	-	-	-	-	-	-	3	-
CO5	-	-	1	-	3	2	-	-	-	3	-	-	-	2
CO6	-	-	-	-	-	-	3	1	-	3	-	2	-	3

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

B. Recommended Study Material:**❖ Reference book:**

1. Books, Magazines ,Journals & online course platforms of related topics

❖ Web material:

1. www.ieeeexplore.ieee.org
2. www.sciencedirect.com
3. www.elsevier.com

4. <https://www.udemy.com/>
5. <https://www.udacity.com/>
6. <https://nptel.ac.in/course.html>
7. <https://www.futurelearn.com/>

❖ **Software:**

1. ASP.NET
2. PYTHON/MATLAB
3. PHP
4. ANDROID/IOS

B. Tech. (Computer Science & Engineering) Programme

SYLLABI **(Semester – 4)**

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CE245: DATA STRUCTURE AND ALGORITHMS

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	4
Marks	100	50	-	150	

Pre-requisite courses:

- Programming Language

A. Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to Data Structure.	04
2.	Linear Data Structure	12
3.	Non Linear Data Structure	16
4.	Sorting	10
5.	Searching	01
6.	Dictionaries	02
	Total hours (Theory) :	45
	Total hours (Lab) :	30
	Total hours :	75

B. Detailed Syllabus:

1.	Introduction	04 Hours	08%
	Introduction to data structure (Types of data structure), Introduction to algorithms. Algorithm Analysis and Big O notation, Memory representation of Array: Row Order and Column Order, Abstract Data Types(ADT)		
2.	Linear Data Structure	12 Hours	27%
	Stack: Operations: push, pop, peep, change, Applications of Stack: Recursion: Recursive Function Tracing, Principles of recursion, Tail recursion, Removal of Recursion, Tower of		

	Hanoi, Conversion: Infix to Postfix, Infix to Prefix. Evaluation: Prefix and Postfix expression, Queue Simple Queue: Insert and Delete operation, Circular Queue: Insert and Delete operation, Concepts of: Priority Queue, Double-ended Queue, Applications of Queue, Linked List: Memory Representation of LL, Singly Linked List, Doubly Linked List, Circular Linked List, Applications of Linked List		
3.	Non Linear Data Structure	16 Hours	36%
	Tree: Tree Concepts, Tree Traversal Techniques: Pre-order, Post-order and In-order (Recursive and Iterative), Binary Search Tree: Iterative and Recursive, Balanced Trees (AVL Trees, Applications of Tree, Heaps: priority queues and Binary Heaps, Graph: Graph concepts, Memory Representation of Graph, BFS and DFS, Applications of Graph		
4.	Sorting	10 Hours	23%
	Sorting (concepts, Selection Sort, Bubble Sort, Merge Sort, Radix Sort, Insertion Sort, Heap Sort, Quick Sort)		
5.	Searching	01 Hours	02%
	Sequential Search, Binary Search		
6.	Dictionaries	02 Hours	04%
	Hashing, Hashing Functions, Collision-Resolution Techniques, Applications		

Course Outcome (COs):

At the end of the course, the students will be able to:

CO1	Understand and Implement Algorithms and core Data Structures such as stack, queue, hash table, priority queue, binary search tree and graph in programming language.
CO2	Analyse data structures in storage, retrieval and computation of ordered or unordered data.
CO3	Compare alternative implementations of data structures with respect to demand

	and performance.
CO4	Describe and evaluate the properties, operations, applications, strengths and weaknesses of different data structures.
CO5	Apply and select the most suitable data structures to solve programming challenges.
CO6	Discover advantages and disadvantages of specific algorithms.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	2	-	-	-	-	-	-	-	-	-	3	-
CO2	-	2	-	-	-	-	-	-	-	-	-	-	3	-
CO3	-	3	3	3	-	-	-	-	-	-	-	-	2	-
CO4	-	1	-	1	2	-	-	-	-	-	-	-	2	-
CO5	2	2	2	2	-	-	-	-	-	-	-	2	3	-
CO6	2	-	-	-	-	-	-	-	-	-	-	-	2	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

C. Recommended Study Material:

❖ Text book:

1. An Introduction to Data Structures with Applications, Jean-Paul Tremblay, Paul G. Sorenson, McGraw-Hill.
2. Data structure with C, Lipschutz, TMH
3. Introduction to Algorithms: Cormen, Leiserson, Rivest and Stein: Prentice Hall of India
4. Data Structures and Algorithms: Aho, Hopcroft and Ullmann: Addison Wesley.

❖ Reference book:

1. Classic Data structures, D.Samanta, Prentice-Hall International.0
2. Data Structures using C & C++, Ten Baum, Prentice-Hall International.

3. Data Structures: A Pseudo-code approach with C, Gilberg & Forouzan, Thomson Learning.
4. Fundamentals of Data Structures in C++, Ellis Horowitz, Sartaj Sahni, Dinesh Mehta, W. H. Freeman.
5. “A Practical Introduction to Data Structures and Algorithm Analysis” by Clifford A. Shaffer
6. Data Structures and Algorithm in Java: Goodrich and Tamassia: John Wiley and Sons.

❖ **Web material:**

1. <http://www.leda-tutorial.org/en/official/ch02s02s03.html>
2. <http://www.leda-tutorial.org/en/official/ch02s02s03.html>
3. <http://www.softpanorama.org/Algorithms/sorting.shtml>

CE246: Database Management System

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	4	4	-	8	6
Marks	100	100	-	200	

Pre-requisite courses:

- Data Structure

A. Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introductory concepts of DBMS	04
2.	Relational Model	06
3.	Entity-Relationship model	07
4.	Formal Relational Query Languages	06
5.	Relational Database Design	09
6.	Transaction & Recovery Management	07
7.	Advanced Transaction Processing	06
8.	Database Security	06
9.	Indexing and Hashing	07
10.	Query Processing & Query Optimization	02
	Total hours (Theory) :	60
	Total hours (Lab) :	60
	Total hours :	120

B. Detailed Syllabus:

1.	Introductory concepts of DBMS	04 Hours	06%
	Introduction and applications of DBMS, Purpose of database, Data Independence, Database System architecture- levels, Mappings, Database users and DBA		
2.	Relational Model	06 Hours	10%
	Structure of Relational Databases, Database Schema,		

	Schema Diagram, Domains , Relations, Relational Query Languages, Relational Operations		
3.	Entity-Relationship model	07 Hours	12%
	Basic concepts, Design process, Constraints, Keys, Design issues, E-R diagrams, Weak Entity Sets, Extended E-R features- Generalization, Specialization, Aggregation, Reduction to E-R database schema		
4.	Formal Relational Query Languages	06 Hours	10%
	The relational Algebra, The Tuple Relational Calculus, The Domain Relational Calculus		
5.	Relational Database design	09 Hours	15%
	Functional Dependency–definition, Trivial and Non-Trivial FD, Closure of FD set, Closure of attributes, Irreducible set of FD, Normalization – 1NF, 2NF,3NF, Decomposition using FD- Dependency Preservation		
6.	Transaction & Recovery Management	07 Hours	12%
	Transaction concepts, Properties of Transactions, Serializability of transactions, Testing for Serializability, System recovery, Two- Phase Commit protocol, Recovery and Atomicity, Log-based recovery, Concurrent executions of transactions and related problems, Locking mechanism, Solution to Concurrency Related Problems, Deadlock, Two-phase locking protocol, Intent locking		
7.	Advanced Transaction Processing	06 Hours	10%
	Transaction-Processing Monitors, Transactional Workflows, Main-Memory Databases, Real-Time Transaction Systems, Long-Duration Transactions		
8.	Database Security	06 Hours	10%
	Views - What are views for?, View retrievals, View updates, Snapshots (a digression), Materialized view, Security – Security and Authentication, authorization in SQL, Data encryption, Missing Information - An overview of the 3VL approach		

9.	Indexing and Hashing	07 Hours	12%
	Basic Concepts, Ordered Indices, B+-Tree Index Files, B+-Tree Extensions, Multiple-Key Access, Static Hashing, Dynamic Hashing, Comparison of Ordered Indexing and Hashing, Bitmap Indices, Index Definition in SQL		
10.	Query Processing & Query Optimization	02 Hours	03%
	Overview, Measures of Query Cost, Selection Operation, Sorting, Join, Evaluation of Expressions, Transformation of relational Expressions, Estimating Statistics of expression results, Query Evaluation plans		

Course Outcomes (COs):

At the end of the course, students will be able to

CO1	Apply the concepts of engineering i.e collecting data, organize the data in the systematic form, arrange the data in a computational way and applying mathematics formation.
CO2	Analyse how data are stored and maintained using data models. Ready to assimilate the concept of data abstraction and design queries using SQL. Identify how data is represented in the relational model and create relations using SQL language
CO3	Identify and evaluate the constructs in the E-R model and issues involved in developing an E-R diagram. Convert an E-R diagram into a relational database schema. Declare and enforce integrity constraints on database using a state-of-art RDBMS.
CO4	Produce aggregate operators to write SQL queries which are not expressible in relational algebra. “More mathematical” notation may apply and also used in research and other venues. Combining these concepts allows production of sophisticated queries.
CO5	Decompose un-normalized tables into normalized compliant tables. Design and implement a normalize database schema for a given problem-domain.
CO6	Compare transactions and their properties with (ACID) and without ACID. Apply locking protocol to ensure isolation. Develop logging technique to ensure atomicity and durability

CO7	Design a logical view which can be used for analytical tasks. Develop practical experience of the design and implement scalable, secure databases.
CO8	Produce strategies to minimise risks of security breaches in a range of network environments and data storage systems. Compute retrieval time and concluding with suitable indexing technique
CO9	Compare and evaluate query execution plan.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	1	2	1	1	-	-	-	1	2	-
CO2	3	3	3	2	3	1	2	-	-	-	-	3	2	1
CO3	3	3	3	3	2	2	1	3	2	1	-	2	3	2
CO4	3	3	1	3	1	1	-	2	2	-	-	2	2	2
CO5	2	2	3	2	3	1	2	2	1	-	-	3	1	1
CO6	3	3	2	1	-	2	-	-	1	-	-	2	-	1
CO7	3	3	3	2	3	1	-	2	2	1	-	3	2	2
CO8	2	3	2	2	2	2	2	3	1	-	-	3	3	1
CO9	2	2	3	1	3	-	-	1	1	-	-	3	3	1

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

C. Recommended Study Material:

❖ Text book:

1. Database System Concepts, Abraham Silberschatz, Henry F. Korth & S. Sudarshan, McGraw Hill.
2. An introduction to Database Systems, C J Date, Addison-Wesley

❖ Reference book:

1. “Fundamentals of Database Systems”, R. Elmasri and S.B. Navathe, the Benjamin / Cumming Pub. Co

2. SQL,PL/SQL the Programming Language of oracle, Ivan Bayross, BPB Publications
3. Oracle: The Complete Reference, George Koch, Kevin Loney, TMH /oracle press

❖ **Web material:**

1. <http://www.sql.org>
2. <http://www.w3schools.com>
3. <http://www.sqlcourse.com>

❖ **Software:**

1. Oracle 10g
2. SQL Lite
3. Live SQL
4. Firebase
5. Squirrel SQL
6. Postgre SQL

CE247: WEB TECHNOLOGIES

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	2	4	-	6	4
Marks	100	100	-	200	

Pre-requisite courses:

- N/A

A. Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to WWW	01
2.	HTML 4.0	02
3.	HTML 5.0	02
4.	Cascading Style Sheet 2.0	02
5.	Cascading Style Sheet 3.0	03
6.	Client side Scripting Language (JavaScript)	05
7.	DOM (Document Object Model)	03
8.	JavaScript Framework (Jquery, AngularJS)	08
9.	Basics of PHP	04
	Total hours (Theory) :	30
	Total hours (Lab) :	60
	Total hours :	90

B. Detailed Syllabus:

1.	Introduction to WWW	01 Hours	03%
	Introduction of Web1.0/Web2.0/3.0, URL, URI, URN, DNS Client/Server Model, Protocols – TCP/IP, HTTP/HTTPS, SMTP, POP3, MIME, IMAP. Website vs. Portal vs. Blog vs. Forum, HTTP Request/Response Header.		
2.	HTML 4.0	02 Hours	07%
	Introduction of HTML, Headings, Comment, Paragraphs,		

	Formatting, Fonts, Links, Images, Tables, Lists, Forms.		
3.	HTML 5.0	02 Hours	07%
	Introducing HTML5, Structural and Semantics Elements, Input types, Form Elements and Attributes, Rich Media: Audio, Video, Canvas and SVG. HTML5 APIs: Geolocation, Drag/Drop, Web Storage, App Cache, Web Workers.		
4.	Cascading Style Sheet 2.0	02 Hours	07%
	CSS Introduction, Types of CSS, CSS Id & Class, CSS Box Model, CSS Properties: Background, Text, Fonts, Links, Lists, Tables, Positioning, Display, Dimension, Floating, Pseudo-class, Pseudo-element, Image Opacity.		
5.	Cascading Style Sheet 3.0	03 Hours	20%
	Introducing CSS3, Selectors, Pseudo-classes and Pseudo-elements Borders and Box Effects, Background Images and Other Decorative Properties, Text Effects and Typographic Styles, Web Fonts, 2D Transforms, 3D Transforms, Transitions and Animations, Multiple Columns, User Interface, Flexible Box Layout and Grid Layout.		
6.	Client side Scripting Language (JavaScript)	05 Hours	17%
	Introduction to JavaScript, Data Types and Variables, Type Conversion, Operators, Decisions, Loops and Functions, Array, Common Mistakes, Debugging and Error Handling, Programming the Browser, Events, HTML Forms: Interacting with User, String Manipulation, Date, Time and Timers, Storing Information: Cookies.		
7.	DOM (Document Object Model)	03 Hours	10%
	DOM Tree, Descending with Child Nodes, Ascending with Child Nodes, Finding an Element by ID, Tag Name and Class, Creating Element, Deleting and Copying Content. Working with DMO: Dynamic Scripts, Dynamic Styles, Manipulating Tables.		
8.	JavaScript Framework (Jquery, AngularJS)	08 Hours	27%
	JQuery: jQuery core, Selectors, Attributes, Events, Effects		

	and jQuery UI. AngularJS: Modules, Directives, Controller, Data Bindings, Scopes, Filter, Table, SQL, Event		
9.	Basics of PHP	04 Hours	12%
	Overview of PHP, Environment Setup, PHP: Array, String, MySql with PHP		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand the concepts, technologies and standards underpinning the World Wide Web, Web 1.0, Web 2.0, Protocols etc.
CO2	Apply a structured approach to identify needs, interests, and functionality of a website using well-structured, easily maintained, standards-compliant, accessible HTML and CSS code.
CO3	Develop dynamic web pages using JavaScript.
CO4	Use scripting languages and web services to transfer data and add interactive components to web pages.
CO5	Design web applications using database technologies to store persistent data.
CO6	Create, build and deploy data-driven web applications for multiple platforms.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	1	2	1	1	-	-	-	2	-	2	-	-	1
CO2	3	3	3	1	3	2	-	2	2	-	2	2	3	2
CO3	3	3	3	2	3	2	-	2	2	-	2	2	3	3
CO4	3	2	3	1	3	2	-	2	2	-	2	2	3	3
CO5	3	2	3	1	3	2	-	1	2	-	2	2	3	3
CO6	1	2	2	1	3	-	-	1	1	-	1	1	1	1

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

C. Recommended Study Material:

❖ Text book:

1. HTML 5 and CSS 3.0 to the Real World by Alexis Goldstein, Sitepoint publication.
2. Beginning JavaScript – 4th Edition by Paul Wilton, Jeremy McPeak, Wrox Publication.
3. Learning JQuery Third Edition, Jonathan Chaffer Karl Swedberg, Packt Publishing.
4. AngularJS O'Reilly, by Shyam Seshadri

❖ Reference book:

1. Learning Web Design by Jennifer Niederst Robbins, 3rd Edition, O'Reilly.
2. HTML5 Web Application Development by Example by J.M Gustafson.
3. Programming HTML5 Application by Zacbary Kessin, O'Reilly
4. JavaScript Bible, Gold Edition by Danny Goodman.
5. JavaScript for Absolute Beginners by Terry McNavage. Apress publication.
6. CSS Cook book By Christopher Schmitt, O'Reilly publication.
7. JQuery Cookbook, O'Reilly Media by Cody Lindley.

❖ Web material:

1. www.w3schools.com
2. www.tutorialspoint.com
3. www.tizag.com

❖ Software:

1. Sublime Text 3
2. Atom (text editor)
3. Visual code

CE248: OPERATING SYSTEM

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	0	5	4
Marks	100	50	0	150	

Pre-requisite courses:

- Introduction to computer and computer architecture.

A. Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction	03
2.	Process Management	06
3.	Inter process Communication	08
4.	Deadlock	06
5.	Memory Management	08
6.	Input Output Management	04
7.	File Systems	06
8.	Unix/Linux File System	04
	Total hours (Theory) :	45
	Total hours (Lab) :	30
	Total hours :	75

B. Detailed Syllabus:

1.	Introduction	03 Hours	06%
	What is an OS? Evolution Of OS, OS Services, Types Of OS, Concepts of OS, Different Views Of OS		
2.	Process Management	06 Hours	10%
	Process, Process Control Block, Process States, Threads, Types of Threads and Dispatching, Concurrent Threads, CPU Scheduling, Protection : Address space, Address Translation		
3.	Inter process Communication	08 Hours	16%

	Race Conditions, Critical Section, Co-operating Thread/ Mutual Exclusion. Hardware Solution, Strict Alternation, and Peterson's Solution. The Producer Consumer Problem, Semaphores, Event Counters, Monitors. Message Passing and Classical IPC Problems: Reader's & Writer Problem, Dining Philosopher Problem.		
4.	Deadlock	06 Hours	16%
	Deadlock Problem, Deadlock Characterization, Deadlock Detection, Deadlock recovery, Deadlock avoidance: Banker's algorithm for single & multiple Resources. Deadlock Prevention.		
5.	Memory Management	08 Hours	20%
	Paging: Principle Of Operation, Page Allocation, H/W Support For Paging. Multiprogramming With Fixed partitions, Segmentation, Swapping. Virtual Memory: Concept, Performance Of Demand Paging, Page Replacement Algorithms, Thrashing and Working Sets.		
6.	Input Output Management	04 Hours	10%
	I/O Devices, Device Controllers, Direct Memory Access. Principles Of Input/output S/W : Goals Of The I/O S/W, Interrupt Handler, Device Driver, Device Independent. I/O Software Disks: RAID levels, Disk Arm Scheduling Algorithm, Error Handling		
7.	File Systems	06 Hours	16%
	File Naming, File Structure, File Types, File Access, File Attributes, File Operations, Memory Mapped Files. Directories: Hierarchical Directory System, Pathnames, Directory Operations. File System Implementation, Contiguous Allocation, Linked List Allocation, Linked List Using Index, Inodes.		
8.	Unix/Linux File System	04 Hours	06%
	Buffer Cache, Inodes, The system calls - ialloc, ifree, namei, alloc and free. Mounting and Unmounting , files systems, Network File systems. EXT file system in linux		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Visualize and understand Operating system functionality and working of OS. Understanding of functionality, services of operating system and differentiate between different types of OS.
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CO2	Define thread and process. Visualize how processes and threads are managed by the operating systems. Simulate and analyse various process scheduling algorithms. Explain and analyse different Inter process communication techniques
CO3	Describe deadlock and classify detection, recovery, prevention and avoidance algorithms. Test scenarios to report deadlock
CO4	Compare and Evaluate various memory management schemes. Simulate and analyse Memory management algorithms. Identify and describe the role of I/O devices.
CO5	Understand the file systems. Understand the secondary storage and simulate disk scheduling algorithm.
CO6	Understand the basic commands of linux file systems

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO 1	3	-	-	-	2	-	-	-	-	-	-	1	-	-
CO 2	3	2	-	-	2	-	-	-	-	-	-	-	1	-
CO 3	3	2	-	-	2	-	-	-	-	-	-	-	1	-
CO 4	3	2	-	-	2	-	-	-	-	-	-	-	1	-
CO 5	3	2	-	-	2	-	-	-	-	-	-	-	1	-
CO 6	3	2	-	-	2	-	-	-	-	-	-	1	1	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

C. Recommended Study Material:

❖ Text book:

1. Modern Operating Systems -By Andrew S. Tanenbaum, Third Edition PHI
2. Operating System Concepts Avi Silberschatz, Peter Baer Galvin, Greg Gagne, Ninth Edition, Wiley

❖ Reference book:

1. Operating Systems, D.M. Dhamdhare, TMH

2. Operating Systems Internals and Design Principles, William Stallings, Seventh Edition, Prentice Hall
3. Unix System Concepts & Applications, Sumitabha Das, TMH
4. Unix Shell Programming, Yashwant Kanitkar, BPB Publications

CE256: COMPUTER ORGANIZATION

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	0	-	3	3
Marks	100	00	-	100	

Pre-requisite courses:

- Digital Electronics

A. Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to digital logic Circuit	03
2.	Register Transfer and Microoperations	08
3.	Basic Computer Organization and Design	08
4.	Central Processing Unit	08
5.	Pipeline and Vector Processing	04
6.	Computer Arithmetic	08
7.	Memory Organization	06
	Total hours (Theory) :	45
	Total hours (Lab) :	00
	Total hours :	45

B. Detailed Syllabus:

1.	Introduction to digital logic Circuit	03 Hours	07%
	Digital Computers, Logic Gates, Combinational Circuits (Half Adder, Full Adder), Flip-Flops(SR, D, JK, T, Edge-Triggered)		
2.	Register Transfer and Microoperations	08 Hours	18%
	Register Transfer Language, Register Transfer, Bus and Memory Transfers; Arithmetic, Logic & Shift Microoperation; Arithmetic Logic Shift Unit		

3.	Basic Computer Organization and Design	08 Hours	18%
	Instruction Codes, Computer Registers, Computer Instructions, Timing and Control, Instruction Cycle, Instruction Cycle, Memory Reference Instructions, Input-Output and Interrupt, Complete Computer Description, Design of Basic Computer, Design of Accumulator Logic		
4.	Central Processing Unit	08 Hours	17%
	Introduction, General Register Organization, Stack Organization, Instruction Formats, Addressing Modes, Data Transfer and Manipulation, Program Control, RISC		
5.	Pipeline and Vector Processing	04 Hours	09%
	Parallel Processing, Pipelining, Arithmetic Pipeline, Instruction Pipeline, RISC Pipeline, Vector Processing, Array Processors		
6.	Computer Arithmetic	08 Hours	18%
	Introduction: Binary, Octal, Decimal, Hexadecimal representation, Integer Numbers: Sign-Magnitude, 1's complement, 2's complement, Addition and Subtraction, Multiplication Algorithm, Multiplication Algorithm, Division Algorithm, Floating Point Representation and Arithmetic Operations		
7.	Memory Organization	06 Hours	13%
	Memory Hierarchy, Main Memory, Auxiliary Memory, Associative Memory, Cache Memory, Virtual Memory		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Recognize elements of digital logic circuit. Moving from design of single bit function to multibit function. (Flip flop, Logic Gates, Combinational Circuit). Design circuit for fixed function arithmetic function. Understand the notation of writing register transfer language
CO2	Design and examine the different Arithmetic, Logic and Shift circuit & Design control unit of Arithmetic, Logic and Shift Circuit

CO3	Differentiate and conceptualize instruction and arithmetic level parallelism & Identify and compare different methods for computer I/O mechanisms.
CO4	Demonstrate and evaluate computer arithmetic operations on integer and real numbers using hardwired algorithm
CO5	Categorize memory organization and understand functioning of internal cache memory hardware.
CO6	Understand and differentiate n-way set associative memory

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	3	3	-	-	-	-	1	1	-	-	1	2	-
CO3	3	3	3	1	-	-	-	1	1	-	-	-	2	-
CO4	2	2	3	-	-	-	-	1	1	-	-	-	2	-
CO5	2	2	-	-	-	-	-	-	-	-	-	-	2	-
CO6	3	3	-	-	-	-	-	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

C. Recommended Study Material:

❖ Text book:

1. Computer System Architecture, Morris Mano (3rd Edition) Prentice Hall.

❖ Reference book:

1. William Stalling, Computer Organization & Architecture-Designing for Performance, Pearson Prentice Hall (8th Edition).
2. A.S. Tananbum, Structured Computer Organization, Pearson Publisher
3. The Essentials of Computer Organization and Architecture Linda Null, Julia Lobur
4. John P Hayes, Computer Architecture & Organization, McGraw-Hill

5. Computer Architecture: Pipelined and Parallel Processor Design Michael J. Flynn (4th edition)

CE255: SOFTWARE GROUP PROJECT-II

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	0	4	-	4	2
Marks	0	100	-	100	

Pre-requisite courses:

- Programming Language, Software Engineering.

A. Outline of the Course:

- Student at the beginning of a semester may be advised by his/her supervisor (s) for recommended courses.
- Students will work together in a team (at most three) with any programming language.
- Students are required to get approval of project definition from the department.
- After approval of project definition students are required to report their project work on weekly basis to the respective internal guide.
- Project will be evaluated at least once per week in laboratory Hours during the semester and final submission will be taken at the end of the semester as a part of continuous evaluation.
- Project work should include whole SDLC of development of software / hardware system as a solution of particular problem by applying principles of Software Engineering.
- Students have to submit project with following listed documents at the time of final submission.
 - d. Final Project Report
 - e. Project Setup file with Source code
 - f. Project Presentation (PPT)
- A student has to produce some useful outcome by conducting experiments or project work.

Total hours (Theory) : 00

Total hours (Lab) : 30

Total hours : 30

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Identify problems present in society by surveying variety of domains and convert in project definition.
CO2	Explore new ideas and techniques to solve it. Create, select and apply appropriate techniques, resources, modern engineering and IT tools to solve problem.
CO3	Correlate knowledge of different subjects and apply theoretical knowledge to implement project for identified problem.
CO4	Apply ethical principles and commit to responsibilities and norms of the project.
CO5	Write technical report and deliver presentation by applying different visualization tools and evaluation metrics.
CO6	Apply engineering and management principles to achieve project goal.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	3	-	2	-	-	-	-	-	-	-	-	-	-
CO2	-	-	3	-	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-	-	3	-
CO4	-	-	-	-	-	-	-	3	2	-	-	-	-	-
CO5	-	-	1	-	3	2	-	-	-	3	-	-	-	-
CO6	3	2	2	1	2	1	-	-	2	3	-	1	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

B. Recommended Study Material:❖ **Reference book:**

1. Books, Magazines ,Journals & online course platforms of related topics

❖ **Web material:**

1. www.ieeexplore.ieee.org
2. www.sciencedirect.com
3. www.elsevier.com

4. <https://www.udemy.com/>
5. <https://www.udacity.com/>
6. <https://nptel.ac.in/course.html>
7. <https://www.futurelearn.com/>

❖ **Software:**

1. ASP.NET
2. PYTHON/MATLAB
3. PHP
4. ANDROID/IOS

B. Tech. (Computer Science & Engineering) Programme

SYLLABI (Semester –5)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CS341: ARTIFICIAL INTELLIGENCE

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr No.	Title of the unit	Minimum number of Hours
1.	Introduction to AI, Problems, Problems Space and Search, Heuristic Techniques	10
2.	Logic and Programming Languages in AI	05
3.	Knowledge Representation and Rules	04
4.	Reasoning	06
5.	Weak Slot-And-Filler Structure and Game Playing and Planning	10
6.	NLP and Text Analytics and Neural Networks	06
7.	Expert Systems and Optimization Techniques and AI & ML Tools	04

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

1. Introduction to AI 10 Hours 20 %

What is AI, Applications of AI, characteristics, advantages and disadvantages.

Problems, Problems Space and Search, Heuristic Search Techniques

Defining The Problems as a State Space Search, Production Systems, Problem Characteristics, Production System Characteristics, Issues In The Design Of Search Programs,

Heuristic Search Techniques:

Hill Climbing, A*, AO*, Simulated Annealing, Branch and Bound, Nearest Neighbour, Blind Search Techniques: DFS, BFS, Best First Search, Control Strategies.

2 Logic and Programming Languages in AI 05 Hours 15 %

Logic:

Propositional Logic, Predicate Logic and Fuzzy Logic, Monotonic and non-Monotonic

Programming Languages:

Introduction to Prolog: Syntax & Numeric Function, Basic List Manipulation Functions In Prolog, Functions, Predicates & Conditional, Input, Output & Local Variables, Iteration & Recursion, Property Lists & Arrays. GUI Version of Prolog.

Python Programming: Syntax, Data Type, Libraries

: NumPy, Numba, NumExpr, SciPy, AstroPy, Pandas, SymPy, Matplotlib, Jupyter, Ipython

3. Knowledge Representation 04Hours 10 %

Knowledge Representation:

Knowledge Representations And Mappings, Approaches To Knowledge Representation.

Representing Knowledge using Rules

Procedural Versus Declarative Knowledge, Logic Programming,

	Forward Versus Backward Reasoning.		
4	Reasoning	06Hours	10 %
	Symbolic Reasoning Under Uncertainty and Statistical Reasoning:		
	Introduction To Non-monotonic Reasoning, Logics For Nonmonotonic Reasoning		
	Statistical Reasoning		
	Probability And Bays' Theorem, Certainty Factors And Rule-Base Systems, Bayesian Networks, Dempster-Shafer Theory		
5.	Weak Slot-And-Filler Structure and Game Playing and Planning	10 Hours	20 %
	Weak Slot-And-Filler Structure:		
	Semantic Nets, Frames, Ontology, OWL, Reasoner		
	Game Playing and Planning:		
	Introduction: Games as Search Problems, Perfect Decisions in Two-Person Games, Imperfect Decisions, Alpha-Beta Pruning, Games That Include anElement of Chance, State-of-the-Art Game Programs : Chess, Checkers or Draughts, Othello, Backgammon, Go		
	The Blocks World, Components Of A Planning System, Goal Stack Planning, Nonlinear Planning Using Constraint Posting, Hierarchical Planning, Reactive Systems.		
6.	NLP and Text Analytics and Neural Networks	06 Hours	15 %
	NLP and Text Analytics:		
	Introduction, Syntactic Processing, Semantic Analysis, Semantic Analysis, Discourse And Pragmatic Processing, Text Analytics, Text pre-processing, Bag of Words, Word Cloud, Machine Translation, sentiment analysis		
	Neural Networks		
	Introduction: Simple Perceptron, Hopfield Network, Learning In Neural Network, Application Of Neural Networks, Recurrent Networks, Deep Neural Network, Convolution Network, Restricted Boltzmann machine, Transfer learning		

7. Expert Systems and Optimization Techniques and AI & ML 04 Hours 10 %
Tools

Expert Systems:

An Introduction To Expert System, Explanation Facilities, Expert System Developments Process, Knowledge Acquisition.

Optimization Techniques and AI & ML Tools:

Genetic Algorithm (GA), Ant Colony Optimization (ACO),

Particle Swarm Optimization(PSO), Honey Bee

AI , Machine Learning and Data Analytics Tools

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand the various searching techniques, constraint satisfaction problem and example problems- game playing techniques.
CO2	Apply these techniques in applications, which involve perception, reasoning and learning.
CO3	Demonstrate fundamental understanding of the history of artificial intelligence (AI) and its foundations.
CO4	Demonstrate awareness and a fundamental understanding of various applications of AI techniques in intelligent agents, expert systems, artificial neural networks and other machine learning models.
CO5	Demonstrate proficiency-developing applications in an 'AI language', expert system shell, or data-mining tool.
CO6	Demonstrate an ability to share in discussions of AI, its current scope and limitations, and societal implications.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	1	1	1	1	1	1	1	-	-	2	2	2	3
CO2	2	2	-	-	-	-	-	1	2	1	1	1	2	-
CO3	3	1	1	-	-	1	1	-	-	-	-	-	2	-
CO4	3	2	3	3	-	-	-	1	2	2	2	2	3	3
CO5	3	2	2	2	3	2	2	-	-	-	-	-	2	-
CO6	2	1	1	-	-	3	-	-	-	-	2	2	3	3

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) If there is no correlation, put “-”

C. Instructional Methods and Pedagogy:

- At the start, of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Faculty would use coached problem-solving method as it is the class format in which faculty provide a structured, guided context for students working collaboratively to solve problems.
- Assignments based on topic content will be given to the students at the end of each unit/topic.
- Surprise tests/Quizzes will be conducted.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Minimum 30 programs of PROLOG in terms of experiments are suggested in the laboratory.

D. Recommended Study Material:

❖ Text Books:

1. “Artificial Intelligence” -By Elaine Rich and Kevin Knight (2nd Edition) Tata Mcgraw-Hill.
2. Stuart J. Russell and Peter Norvig, Artificial Intelligence 3e: A Modern Approach, 3rd Edition. Person
3. Introduction to Prolog Programming By Carl Townsend

❖ Reference Books:

1. “Artificial Intelligence and Expert System, Development”-By D.W. Rolston, Mcgraw-Hill International Edition.
2. “Artificial Intelligence And Expert Systems ”By D.W. Patterson
3. “PROLOG Programming For Artificial Intelligence” By Ivan Bratko, Addison-Wesley
4. “Programming with PROLOG” –By Klocksin and Mellish.

CE341: MICROPROCESSOR ARCHITECTURES AND ASSEMBLY PROGRAMMING

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	4
Marks	100	50	-	150	

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	The Processor: 8086	06
2.	8086 Instruction set and Assembler Directives	06
3.	The Art of Assembly Language Programming with 8086	06
4.	Special Architectural features and related Programming	06
5.	DMA Controllers, Multi-microprocessor systems	03
6.	80286 Processor	07
7.	80386 Processor	07
8.	Current Era of Microprocessors	04

Total hours (Theory): 45

Total hours (Lab): 30

Total hours: 75

B. Detailed Syllabus:

- | | | |
|--|-----------------|------------|
| 1. THE PROCESSOR: 8086 | 05 Hours | 10% |
| Register Organisation of 8086, Architecture, Signal Descriptions, Physical Memory Organisation, General Bus Operation, I/O Addressing Capability, Special Purpose Activities, Minimum & Maximum Modes 8086 System and Timings. | | |
| 2. 8086 INSTRUCTION SET AND ASSEMBLER DIRECTIVES | 10 Hours | 20% |
| Machine Language Instruction Formats, Addressing Modes, Instruction Set, Assembler Directives and Operators | | |
| 3. THE ART OF ASSEMBLY LANGUAGE PROGRAMMING WITH 8086 | 15 Hours | 30% |
| A Few Machine Level Programs, Machine Coding the Programs, Programming with an Assembler, Assembly Language Example Programs | | |
| 4. SPECIAL ARCHITECTURAL FEATURES AND RELATED PROGRAMMING | 10 Hours | 20% |
| Introduction to Stack, Stack Structure, Interrupt and Interrupt Service Routines, Interrupt Cycle of 8086, Non-Maskable Interrupt, Maskable Interrupt, Interrupt Programming, MACROS, Timing and Delays. | | |
| 5. DMA CONTROLLER, MULTIMICROPROCESSOR SYSTEMS | 10 Hours | 20% |
| DMA Controller, Interconnection Topologies, Software Aspects of Multi-microprocessor Systems, Tightly Coupled and Loosely Coupled Systems | | |
| 6. 80286 PROCESSOR | 07 Hours | 17% |
| Salient Features of 80286, Internal Architecture of 80286, Signal Description of 80286, Real Addressing Mode, Protected Virtual Address Mode (PVAM) | | |
| 7. 80386 PROCESSOR | 07 Hours | 17% |
| Salient Features of 80386DX, Architectural and Signal Descriptions of 80386, Register Organisation of 80386, Addressing Modes, Data Types of 80386, Real Address Mode of 80386, Protected Mode of 80386, Segmentation. | | |
| 8. CURRENT ERA OF MICROPROCESSORS | 04 Hours | 06% |
| Core i3, i5, i7, xen Processor, multi core processors. | | |

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Recognize elements of digital logic circuit. Moving from design of single bit function to multibit function. (Flip flop, Logic Gates, Combinational Circuit). Design circuit for fixed function arithmetic function. Understand the notation of writing register transfer language.
CO2	Design and examine the different Arithmetic, Logic and Shift circuit & Design control unit of Arithmetic, Logic and Shift Circuit.
CO3	Differentiate and conceptualize instruction and arithmetic level parallelism & Identify and compare different methods for computer I/O mechanisms.
CO4	Demonstrate and evaluate computer arithmetic operations on integer and real numbers using hardwired algorithm.
CO5	Categorize memory organization and understand functioning of internal cache memory hardware.
CO6	Understand and differentiate n-way set associative memory.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	3	3	-	-	-	-	1	1	-	-	1	2	-
CO3	3	3	3	1	-	-	-	1	1	-	-	-	2	-
CO4	2	2	3	-	-	-	-	1	1	-	-	-	2	-
CO5	2	2	-	-	-	-	-	-	-	-	-	-	2	-
CO6	3	3	-	-	-	-	-	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) If

there is no correlation, put “-”

C. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.

- Attendance is compulsory in lectures and laboratory which carries 5 Marksweightage.
- Two internal exams will be conducted and average of the same will be converted to equivalent of 15 Marks as a part of internal theory evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval. It carries a weightage of 5 Marks as a part of internal theory evaluation.
- Surprise tests/Quizzes/Seminar will be conducted which carries 5 Marks as a part of internal theoryevaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.

D. Recommended Study Material:

❖ Text Books:

1. “Microprocessors and Interfacing Programming and Hardware”, DouglasVHall, McGraw-Hill Education India Pvt. Ltd.
2. “Advanced Microprocessor and Peripherals –Architecture, Programming and Interfacing”, A.K.Ray & K.M Bhurchandi, Tata Mc Graw Hill, 2006.
3. “8086 Programming and Advance Processor Architecture”, M. T. Savaliya, Wiley-India.

❖ Reference Books:

1. “Microcomputer systems: The 8086 / 8088 Family architecture, Programming and Design”, Yncheng Liu,Glenn A.Gibson, second edition, Prentice Hall of India , 2006
2. The 8088 and 8086 Microprocessors: Programming, Interfacing and Applications, Walter A.Triebel, Avtar Singh, Prentice-Hall of India Pvt. Ltd.
3. “IBM PC Assembly language and programming”, Peter Abel, fifth edition, Pearson education / Prentice Hall of India Pvt. Ltd, 2007.

CE342: DESIGN & ANALYSIS OF ALGORITHMS

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	4
Marks	100	50	-	150	

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	To derive time and space complexity of algorithm.	03
2.	Analysis of Algorithm	06
3.	Greedy Algorithm	07
4.	Divide and Conquer Algorithm	07
5.	Dynamic Programming	08
6.	Exploring Graphs	04
7.	Backtracking & Branch & Bound	05
8.	String Matching and Introduction to NP- Completeness	05

Total hours (Theory): 45

Total hours (Lab): 30

Total hours: 75

B. Detailed Syllabus:

1. Basics of Algorithms and Mathematics	03 Hours	05 %
1.1 What is an algorithm?		
1.2 Performance Analysis, Model for Analysis - Random Access Machine (RAM), Primitive Operations		
1.3 Time Complexity and Space Complexity		
2. Analysis of Algorithm	06 Hours	14 %
2.1 The efficiency of algorithm, average and worst case analysis, elementary operation		
2.2 Asymptotic Notation		
2.3 Analyzing control statement		
2.4 Analyzing Algorithm using Barometer		
2.5 Solving recurrence Equation		
2.6 Sorting Algorithm		
3. Greedy Algorithm	07 Hours	16 %
3.1 General Characteristics of greedy algorithms		
3.2 Problem solving using Greedy algorithm		
3.3 Making change problem		
3.4 Graphs: Minimum Spanning trees (Kruskal's algorithm, Prim's algorithm)		
3.5 Graphs: Shortest paths; The Knapsack Problem; Job Scheduling Problem		
4. Divide and Conquer Algorithm	07 Hours	16 %
4.1 Multiplying large Integers Problem		
4.2 Binary Search		
4.3 Sorting (Merge Sort, Quick Sort)		
4.4 Matrix Multiplication		
4.5 Exponential		

5. Dynamic Programming	08 Hours	18 %
5.1 Introduction, The Principle of Optimality		
5.2 Problem Solving using Dynamic Programming – Calculating the Binomial Coefficient		
5.3 Making Change Problem		
5.4 Assembly Line-Scheduling		
5.4 Knapsack Problem		
5.5 Shortest Path		
5.6 Matrix Chain Multiplication		
5.7 Longest Common Subsequence		
6. Exploring Graphs & Backtracking	04 Hours	09 %
6.1 An introduction using graphs and games,		
6.2 Traversing Trees – Preconditioning Depth First Search- Undirected Graph; Directed Graph, Breath First Search, Applications of BFS & DFS		
7. Backtracking & Branch & Bound	05 Hours	12%
7.1 Backtracking –The Knapsack Problem; The Eight queens problem, General Template		
7.2 Branch and Bound –The Assignment Problem; TheKnapsack Problem, The min-max principle		
8. String Matching and Introduction to NP-Completeness	05 Hours	10%
8.1 The naïve string-matching algorithm		
8.2 The Rabin-Karp algorithm		
8.3 The class P and NP Problems		
8.4 Polynomial reduction		
8.5 NP- Completeness Problem		
8.6 NP-Hard problems		

C. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.

- It deals with guided discovery problems through intriguing puzzles to solve, structured hands on activities, carefully worded leading questions, crucial hints, guided discovery problems escort students step by step through process of scientific discovery.
- Attendance is compulsory in lectures and laboratory which carries a 5% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval. It carries a weight-age of 5%.
- Surprise tests/Quizzes will be conducted which carries 5% component of the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Minimum 10 experiments are suggested in the laboratory related to course content.

D. Student Learning Outcomes:

Upon completion of this course, students will be able to do the following:

- Students will be able to develop efficient and effective computer algorithm. This will help for development of high-quality software and problem-solving approach.
- Students will get confidence for programming and problem-solving methodology.

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Analyse the asymptotic performance of algorithms.
CO2	Derive time and space complexity of different sorting algorithms and compare them to choose application specific efficient algorithm.
CO3	Understand and analyse the problem to apply design technique from divide and conquer, dynamic programming, backtracking, branch and bound techniques and understand how the choice of algorithm design methods impact the performance of programs.
CO4	Understand and apply various graph algorithms for finding shortest path and minimum spanning tree.
CO5	Synthesize efficient algorithms in common engineering design situations.
CO6	Understand the notations of P, NP, NP-Complete and NP-Hard.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	1	-	-	-	-	-	-	-	-	-	-	1	-
CO2	2	2	-	-	-	-	-	-	-	-	-	2	2	-
CO3	3	3	3	3	2	-	-	-	-	-	-	2	2	-
CO4	2	3	3	1	-	-	-	-	-	-	-	-	2	-
CO5	1	-	1	-	-	-	-	-	-	-	-	2	1	1
CO6	3	1	-	-	-	-	-	-	-	-	-	-	1	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) If

there is no correlation, put “-”

E. Recommended Study Material:

❖ Text Books:

1. Introduction to Algorithms by Thomas H. Cormen, Charles E. Leiserson, Ronald Rivest and Clifford Stein, MIT Press

❖ Reference Books:

1. Fundamental of Algorithms by Gills Brassard, Paul Bratley, Pentice Hall of India.
2. Fundamental of Computer Algorithms by Ellis Horowitz, Sartazsahni and sanguthevar Rajasekarm, Computer Sci.P.
3. Design & Analysis of Algorithms by P H Dave & H B Dave, Pearson Education.

❖ Web Materials:

1. <http://www.stanford.edu/class/cs161/>
2. <http://www.itl.nist.gov/div897/sqg/dads/>
3. <http://highered.mcgraw-hill.com/sites/0073523402/>

CE344: COMPUTER NETWORKS

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	4
Marks	100	50	-	150	

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Introduction to Computer Networks	04
2	Data Link Layer	08
3	Medium Access Control Sub Layer	10
4	Network Layer	12
5	Transport Layer	08
6	Application Layer	03

Total hours (Theory): 45

Total hours (Lab): 30

Total hours: 75

B. Detailed Syllabus:

1	Introduction to Computer Networks	04 Hours	09%
1.1	Uses of computer network		
1.2	network hardware, network software		
1.3	OSI model, TCP/IP model, Comparison of OSI and TCP/IP model		
1.4	Example network		
2	Data Link Layer	08 Hours	18%
2.1	Design Issues		
2.2	framing, error control, flow control		
2.3	Error detection and correction		
2.4	Elementary data link protocols		
2.5	simplex, stop and wait, sliding window protocol, HDLC		
3	Medium Access Control Sub Layer	10 Hours	22%
3.1	The channel allocation problem, Multiple Access protocols: ALOHA, CSMA, Collision Free Protocols,		
3.2	Limited Contention Protocols, Wavelength Division Multiple Access Protocols		
3.3	Wireless LAN protocols; Ethernet: Traditional Ethernet, Switched Ethernet, Fast Ethernet, Gigabit Ethernet, IEEE 802.2: LLC, Data link layer switching		
4	Network Layer	12 Hours	27%
4.1	Implementation of connection oriented and connection less service, Comparison of virtual circuit and datagram subnets, Routing algorithms		
4.2	Shortest path routing, Flooding, Distance vector routing, Link state routing, Hierarchical routing, Broadcast routing, Multicast routing, Routing for mobile host		
4.3	Routing in ad hoc network, Congestion control algorithms principles, Prevention policies		
4.4	Congestion control in virtual circuit subnets, Congestion control in		

- datagram subnets, Load shedding,
- 4.5 virtual circuit, Connectionless internetworking, Tunneling,
Internetwork routing and fragmentation
- 4.6 The network layer in the internet: The IP protocol, IPaddresses
- 4.7 Internet control protocol, OSPF, BGP
- 5 Transport Layer 08 Hours 18%**
- 5.1 The transport service: Services provided to the upper layers
- 5.2 Transport service primitives, Socket elements of transport protocols
addressing
- 5.3 Connection establishment, Connection release, Flow control
- 5.4 Multiplexing, Crash recovery the transport protocol: UDP, TCP
- 6 Application Layer 03 Hours 06%**
- 6.1 DNS: The DNS name space, Resource records, Nameservers
- 6.2 Electronic mail: Architecture and services
- 6.3 World Wide Web: Architectural overview, HTTP.

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Analyze layered network architecture and passage of data over communication links
CO2	Analyze delay models in Data Networks using Queueing Systems for messaging and delay sensitive applications
CO3	Design and analyze routing algorithms for Internet and multi-hop autonomous networks
CO4	Analyze flow and rate control algorithms between a sender and receiver in wide area networks
CO5	Apply the network fundamentals to analyze performance.
CO6	Use key networking algorithms in simulation.

Course Articulation Matrix:

	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	1	3	1	-	-	-	-	-	-	-	2	-
CO2	3	3	1	3	1	-	-	-	-	-	-	-	1	-
CO3	3	3	1	3	1	-	-	-	-	-	-	-	1	-
CO4	3	3	1	3	1	-	-	-	-	-	-	-	1	-
CO5	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO6	-	-	-	-	3	-	-	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) If

there is no correlation, put “-”

C. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries 5 Marks weightage.
- Two internal exams will be conducted and average of the same will be converted to equivalent of 15 Marks as a part of internal theory evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval. It carries a weightage of 5 Marks as a part of internal theory evaluation.
- Surprise tests/Quizzes/Seminar will be conducted which carries 5 Marks as a part of internal theory evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.

D. Recommended Study Material:

❖ **Text Books:**

1. Computer Network, Andrew S. Tanenbaum, Prentice Hall PTR

❖ **Reference Books:**

1. Introduction to Data Communication and Networking by Behrouz Forouzan, McGraw Hill
2. Data and Computer Communications, William Stallings, Prentice Hall

❖ **Reference Books:**

1. <http://www.cisco.com>
2. <http://compnetworking.about.com>

CS376: ADVANCED JAVA PROGRAMMING (PE-I)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	2	4	-	6	4
Marks	100	100	-	200	

A. Outline of the Course:

Sr. No.	Title of the unit	Minimum number of Hours
1.	JDBC Programming	05
2.	Servlet	06
3.	Java Server Pages	05
4.	Java Server Faces 2.0	03
5.	Hibernate4.0	05
6.	JAVA web Frameworks: Spring MVC	06

Total Hours (Theory): 30

Total Hours (Lab): 60

Total Hours: 90

B. Detailed Syllabus:

- | | | | |
|----------|--|-----------------|------------|
| 1 | JDBC Programming | 05 Hours | 16% |
| 1.1 | The JDBC Connectivity Model | | |
| 1.2 | Database Programming: Connecting to the Database, Creating a SQL Query, Getting the Results, Updating Database Data | | |
| 1.3 | Error Checking and the SQLException Class, The SQLWarning Class | | |
| 1.4 | The Statement Interface, The ResultSet Interface, Updatable Result Sets | | |
| 1.5 | Executing SQL Queries, ResultSetMetaData, Executing SQL Updates, Transaction Management. | | |
| 2 | Servlet | 06 Hours | 20% |
| 2.1 | Servlet Model: Overview of Servlet, Servlet Life Cycle, HTTP Methods | | |
| 2.2 | Structure and Deployment descriptor | | |
| 2.3 | ServletContext and ServletConfig interface, Attributes in Servlet, Request Dispatcher interface. | | |
| 2.4 | The Filter API: Filter, FilterChain, Filter Config. | | |
| 2.5 | Cookies and Session Management: Understanding state and session, Understanding Session Timeout and Session Tracking, URL Rewriting. | | |
| 3 | Java Server Pages | 05 Hours | 17% |
| 3.1 | JSP Overview: The Problem with Servlets, Life Cycle of JSP Page, JSP Processing, JSP Application Design with MVC, Setting Up the JSP Environment | | |
| 3.2 | JSP Directives, JSP Action, JSP Implicit Objects | | |
| 3.3 | JSP Form Processing, JSP Session and Cookies Handling, JSP Session Tracking | | |
| 3.4 | JSP Database Access, JSP Standard Tag Libraries, JSP Custom Tag, JSP Expression Language, JSP Exception Handling, JSP XML Processing. | | |
| 4 | Java Server Faces2.0 | 03 Hours | 10% |

- 4.1 Introduction to JSF, JSF request processing Life cycle,JSF Expression Language
- 4.2 JSF Standard Component, JSF Facelets Tag, JSFConvertor Tag, JSF Validation Tag
- 4.3 JSF Event Handling and Database Access, JSF Librarie: PrimeFaces

5 Hibernate

05 Hours 17%

- 5.1 Overview of Hibernate, Hibernate Architecture
- 5.2 Hibernate Mapping Types, Hibernate O/R Mapping, and Hibernate Annotation.
- 5.3 Hibernate Query Language

6 Java Web Frameworks: Spring MVC

06 Hours 20%

- 6.1 Overview of Spring, Spring Architecture, bean life cycle,
- 6.2 XML Configuration on Spring, Aspect – orientedSpring
- 6.3 Managing Database, Managing Transaction

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Illustrate database access and details for managing information using the JDBC API
CO2	Design to map java classes and object associations to relational database tables with hibernate framework.
CO3	Demonstrate web application lifecycle using servlet, and JSP.
CO4	Identify the user interface control and build custom tag library using JSF
CO5	Develop real-world applications using java frameworks and understands about MVC architecture with its application.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	2	2	1	2	-	-	-	-	-	-	-	1	-
CO2	3	2	3	1	3	-	-	-	-	-	-	-	2	-
CO3	2	2	2	2	3	-	-	-	-	-	-	-	-	-
CO4	1	2	2	1	3	-	-	-	-	-	-	-	1	-
CO5	2	3	3	2	3	-	-	-	-	-	-	-	3	1

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) If

there is no correlation, put “-”

C. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries 5 Marks weightage.
- Two internal exams will be conducted and average of the same will be converted to equivalent of 15 Marks as a part of internal theory evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval. It carries a weightage of 5 Marks as a part of internal theory evaluation.
- Surprise tests/Quizzes/Seminar will be conducted which carries 5 Marks as a part of internal theory evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.

D. Recommended Study Material:

❖ Text Books:

1. SCWCD, Matthew Scarpino, Hanumant Deshmukh, Jignesh Malavie, Manning publication
2. Hibernate 2nd edition, Jeff Linwood and Dave Minter, Beginning Après publication
3. Java Persistence with MyBatis 3, K. Siva Prasad Reddy, PACKT publication
4. Spring in Action 3rd edition, Craig walls, Manning Publication
5. Java Server Faces in Action, Kito D. Mann, Manning Publication

❖ **Reference Books:**

1. JDBC™ API Tutorial and Reference, Third Edition, Maydene Fisher, JonEllis, Jonathan Bruce, Addison Wesley
2. Beginning JSP, JSF and Tomcat, Giulio Zambon, Apress
3. JSF2.0 CookBook, Anghel Leonard, PACKTpublication

❖ **Web Materials:**

1. http://www.servicearchitecture.com/applicationservers/articles/j2ee_web_site_architecture.html
2. <http://www.oracle.com/technetwork/java/javaee/overview/index.html>
3. <http://www.roseindia.net/struts/hibernate-spring/index.shtml>
4. <http://www.roseindia.net/jsf/>

CS371: DIGITAL IMAGE PROCESSING(PE-I)

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	2	4	6	4
Marks	100	100	200	

A. Outline of the Course:

Sr. No.	Title of the unit	Minimum Number of Hours
1.	Digital Image Fundamentals	02
2.	Spatial Domain Image Enhancement Techniques	04
3.	Frequency Domain Image Enhancement Techniques	03
4.	Image Restoration	03
5.	Colour Image Processing	01
6.	Image Compression	02
7.	Image Segmentation	04
8.	Morphological Image Processing	04
9.	Representation & Description	02
10.	Advanced Research Topics in Image Processing	04

Total hours (Theory): 45Hrs.

Total hours (Lab): 30 Hrs.

Total hours: 75 Hrs.

B. Detailed Syllabus:

- | | | | |
|----------|---|-----------------|------------|
| 1 | Digital Image Fundamentals | 02 Hours | 11% |
| | Image Basics - Elements of visual perception – Human Eye Structure- Why digital images -The digital camera-Data types and 2d representation of digital images- Application fields of Image Processing- Image Acquisition Techniques-Single Sensor-Strip-Array Sensor - Image sampling and quantization - Discrete sampling model Quantization -Noise processes- Spatial & Grey level resolution- Image attributes- Image Types- Basic relationship between pixels-Neighbourhood- Adjacency- Distance measures | | |
| 2 | Spatial Domain Image Enhancement Techniques | 04Hours | 13% |
| | Intensity based transforms- power law, log ,image negative, Histogram based transforms- Histogram processing, Equalization, Local Enhancement, Specification, Image Averaging, Subtraction, AND-OR-NOT Operations between images, Smoothing Filters- Linear Filters, Order Statistic Filters, Sharpening Filters-Laplacian Filter , Unsharp masking , high boost filtering | | |
| 3 | Frequency Domain Image Enhancement Techniques | 03 Hours | 9% |
| | Introduction to signal-frequency concept, Discrete Fourier Transform and Inverse Transform, properties of Fourier Transform ,1D Fourier Transform, 2D Fourier Transform, Frequency Filtering Concepts, Smoothing Filters(Low pass Filters)- Ideal low-pass, Butterworth low-pass, Gaussian low-pass filters, Sharpening Filters(High-Pass filters)- Ideal , Butterworth, Gaussian, Laplacian Filter in Frequency domain, Unsharp masking , high boost filtering ,Homomorphic Filtering , Convolution & Correlation Theorem , Fast Fourier Transform | | |
| 4 | Image Restoration | 03 Hours | 11% |

Image Degradation-Restoration Model, Noise Models, Noise probability density functions, Noise estimation parameters, Spatial Domain Restoration- Mean filters – Arithmetic mean, Geometric mean, Harmonic mean Order Statistics Filters- Median filter, Max & min filter, Midpoint filter, Alphatrimmed filter, Adaptive filters, Adaptive local noise filter, Adaptive median filter, Frequency Domain Restoration Techniques- Band reject filter, Band-pass filter, Notch filter, Optimum Notch filter, Estimation of Degradation function-By experiment, By Modeling , Inverse Filtering, WienerFiltering, Constrained Least Square Filtering, Geometric Mean filter

5 Colour Image Processing

**01 5%
Hours**

Colour fundamentals , Colour Models, HIS,YIQ,RGB,CMYK,CIE Lab, XYZ, Intensity Slicing, Grey level to Colour Transform, Tone & Colour Correction, Histogram Processing , Smoothing, Sharpening, Segmentation

6 Image Compression

02 Hours 7%

Fundamental to Data Compression, Information Theory, Rate Distortion Theory, Redundancy- Coding, Inter-pixel, psycho- visual redundancy, Fidelity, Image Compression Models, Lossless Compression theorem- Huffman, Arithmetic Coding , LZW, Bit-Plane Coding, Run Length Encoding, Contour Coding, Lossless predictive Coding , Lossy predictive Coding- Transform Coding, DCT, KLT, Wavelet, JPEG Compression, JPEG2000.

7 Image Segmentation

04 Hours 13%

Detection of Discontinuities- Point Detection - LineDetection
- Edge Detection - Edge Linking and Boundary Detection-
Local Processing - Global Processing via the Hough

Transform - Global Processing via Graph-Theoretic Techniques
 - Thresholding - Foundation - The Role of Illumination - Basic
 Global Thresholding - Basic Adaptive Thresholding - Optimal
 Global and Adaptive Thresholding - Use of Boundary
 Characteristics for Histogram Improvement and Local
 Thresholding - Thresholds Based on Several Variables
 Region-Based Segmentation - Basic Formulation - Region Growing -
 Region Splitting and Merging - Segmentation by Morphological
 Watersheds - Basic Concepts
 -Dam Construction - Watershed Segmentation Algorithm -
 The Use of Markers

8 Morphological Image Processing 04 Hours 13%

Preliminaries: Some Basic Concepts from Set Theory - Logic
 Operations Involving Binary Images - Dilation and Erosion -
 Opening and Closing - The Hit-or-Miss Transformation- Some
 Basic Morphological Algorithms: Boundary Extraction- Region
 Filling- Extraction of Connected Components -
 Convex Hull - Thinning - Thickening - Skeletons – Pruning

9 Representation & Description 02 Hours 7%

Chain Code, Polygonal Approximation, Signatures, Boundary
 segments, Skeletons, Shape numbers, Fourier Descriptors,
 Statistical moments, Regional Descriptors: Topological,
 Texture, Moments of 2D functions, Using Principle
 Components

10 Advance Research Topics In Image Processing 04 Hours 11%

Case Study on the following applications:
 Digital Water Marking, Biometric authentication (face,
 fingerprint, signature recognition), Vehicle number plate
 detection and recognition, Content Based Image Retrieval,
 Text Compression

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Examine, explain and analyse different types of images, general terminology and fundamental concepts of digital image processing.
CO2	Interpret, compare and evaluate the techniques for image enhancement, restoration, segmentation, morphological operation, representation and description.
CO3	Interpret different colour models and apply different image processing algorithms on colour images.
CO4	Categorize and analyse various compression techniques.
CO5	Understand real time problems of image processing field and apply image processing knowledge to solve those problem.

Course Articulation Matrix:

	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11	PO12	PSO1	PSO2
CO1	1	2	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	-	2	2	2	-	-	-	-	-	-	-	-	1
CO3	3	-	2	2	2	-	-	-	-	-	-	-	-	1
CO4	3	-	2	2	2	-	-	-	-	-	-	1	-	1
CO5	3	3	3	3	3	3	2	-	-	-	-	1	2	1

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) If

there is no correlation, put “-”

C. Instructional Method and Pedagogy:

- Lectures will be taken in class room with the use of multi-media presentations and black board – mix of both.
- Assignments based on above course content will be given to the students at the end of each chapter. Each assignment contains minimum 5 questions.
- Quizzes and Surprise tests will be conducted for testing the knowledge of students for particular topic.

D. Recommended Study Material:

❖ Text Books:

1. “Digital Image Processing”, Rafael C Gonzalez, Richard E Woods, 2nd Edition, Pearson Education 2003.
2. “Digital Image Processing”, W. Pratt, Wiley Publication, Fourth Edition, 2013.

❖ Reference Books:

1. “Fundamentals of Digital Image Processing”, A.K. Jain, PHI, New Delhi (1995).
2. “Digital Image Processing Using MATLAB”, Gonzalez/Woods/Eddins , 2nd edition
3. “Digital Image Processing and Computer Vision”. Milan Sonka, Thomson publication, Second Edition.
4. “Digital Image Processing”, S.Jayaraman, S Esakkirajan and T Veerakumar, McGraw Hill Education (India) Private Limited, New Delhi, 2009
5. “Digital Image Processing”, S.Sridhar, Oxford University Press, New Delhi, 2011.

❖ Web Materials:

1. <http://www.imageprocessingplace.com/>

CS372: ADVANCED WEB TECHNOLOGY (PE-I)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	2	4	-	6	4
Marks	100	100	-	200	

A. Outline of the Course:

Sr No.	Title of the unit	Minimum number of Hours
1	Web Server	01
2	Server Side Language (PHP)-Part1	08
3	Sever Side Language (Object Oriented PHP)- Part-2	06
4	MySQL	05
5	Reactive Programming	02
6	Design Pattern	02
7	PHP Web services	02
8	PHP Framework	04

Total Hours (Theory): 30

Total Hours (Lab): 60

Total Hours: 90

B. Detailed Syllabus:

- | | | |
|--|----------------|------------|
| 1 Web Server. | 01Hours | 3% |
| Introduction, Apache Web server, IIS6.0/7.0, WAMP, XAMPP | | |
| 2. Server Side Language (PHP) Part -1 | 08Hours | 27% |
| Embedded PHP in HTML, Adding Dynamic content, Accessing form variable, Understanding Identifier, Operator, Decision and loops, Function, Array and String manipulation, Preserving state with Query Strings, Session and cookies, Working with files and directories, Regular expression, Error and Exception handling | | |
| 3. Server Side Language (Object oriented PHP)-Part 2 | 06Hours | 21% |
| Key OOP concepts: Classes, Objects, Properties, Constant, Constructors and Destructors, Static Class Members. Advanced OOP concepts: Interface, Abstract Class, Object Cloning, Auto loading Class and Namespace, Magic Methods, Optimizing for High Performance | | |
| 4. MySQL | 05Hours | 18% |
| Introduction, Installation, Administration, PHP Syntax, Connection, Create Database, Drop Database, Select Database, Data Types, Create Tables, Drop Tables, Insert Query, Select Query, Where Clause, Update Query, Delete Query, Like Clause Sorting Results ,Using Join NULL Values , Transactions ,Alter Command Indexes ,Temporary Tables ,Clone Tables ,Database Info ,Using Sequences, Handling Duplicates, Database Export, Database Import, MySQL useful Function | | |
| 5. Reactive Programming | 02Hours | 06% |
| Similarities with event-driven programming, Using RxPHP, Installing RxPHP, Observable and observer ,Subject Operator, Writing custom operators , Non-blocking IO , Using React ,Installing React ,React event loop ,Observables and event loop | | |

6. Design Patterns

02Hours

06%

Base patterns: The registry pattern

Creational patterns: The singleton pattern, The prototype pattern, The abstract factory pattern, The builder pattern, The object pool pattern, Behavioral patterns: The strategy pattern, The observer pattern, The lazy initialization pattern, The chain of responsibility pattern, Structural patterns: The decorator pattern

7. PHP Web service

02Hours

06%

RPC and SOAP Services: RPC, SOAP: WSDL, PHP SOAP Client and Server. REST: RESTful URLs, HTTP Features in REST: Create Resources, Read Records, Update and Delete Records.

8. PHP Frameworks

04 Hours

13%

Introduction for Laravel: Installation, Configuration and Directory structure. Architecture Concepts: Request Lifecycle, Service Container, Service Providers, Facades, Contracts Database: Getting Started, Query Builder, Pagination, Migrations Eloquent ORM: Getting Started, Relationships, Collections, Mutators

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	How to use Express JS framework with Node.js with real projects with request
CO2	Handle Form, session and cookies.
CO3	Render Client-side/Server-side website using template engine (Apply)
CO4	Create CRUD operations with MongoDB and Mongoose (Create)
CO5	Build a fast, scalable, feature-rich RESTful API (includes filters, sorts, pagination, and much more) (Create)
CO6	Build a complete, beautiful & real-world application from start to finish (API and server-side rendered website)

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	1	3	2	3	1	-	2	1	1	3	3	1	1
CO2	-	1	3	2	3	1	-	1	1	1	3	3	1	1
CO3	-	1	3	2	3	1	-	1	2	2	3	3	2	1
CO4	-	1	3	2	3	2	-	1	2	2	3	3	3	1
CO5	-	1	3	2	3	2	-	1	2	3	3	3	3	1
CO6	-	1	3	2	3	3	-	2	3	3	3	3	3	2

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) If

there is no correlation, put “-”

C. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries 5 Marks weightage.
- Two internal exams will be conducted and average of the same will be converted to equivalent of 15 Marks as a part of internal theory evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval. It carries a weightage of 5 Marks as a part of internal theory evaluation.
- Surprise tests/Quizzes/Seminar will be conducted which carries 5 Marks as a part of internal theory evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.

D. Recommended Study Material:

❖ **Text Books:**

1. Mastering PHP 7 Branko Ajzele, Packt Pub
2. Restful Web Services, Leonard Richardson, Sam Ruby, David Heinemeier Hansson, O'Reilly.

❖ **Reference Books:**

1. PHP Reactive Programming, Martin Sikora , PacktPub
2. Restful Web Services Cookbook: Solutions for Improving Scalability and Simplicity, Subbu Allamaraju, O'Reilly.

❖ **Web Materials:**

1. www.w3schools.com
2. www.learnphp-tutorial.com

CS348: SOFTWARE GROUP PROJECT -III

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	0	2	-	2	2
Marks	0	100	-	100	

A. Outline of the Courses:

- Student at the beginning of a semester may be advised by his/her supervisor (s) for recommended courses.
- Students will work together in a team (at most **three**) with any programming language.
- Students are required to get approval of project definition from the department.
- After approval of project definition students are required to report their project work on weekly basis to the respective internal guide.
- Project will be evaluated at least once per week in laboratory hours during the semester and final submission will be taken at the end of the semester as a part of continuous evaluation.
- Project work should include whole SDLC of development of software / hardware system as a solution of particular problem by applying principles of Software Engineering.
- Students have to submit project with following listed documents at the time of final submission.
 - a. Project Synopsis
 - b. Software Requirement Specification
 - c. SPMP
 - d. Final Project Report
 - e. Project Setup file with Source code
 - f. Project Presentation (PPT)
- A student has to produce some useful outcome by conducting experiments or project work.

Total hours (Theory): 00

Total hours (Lab): 30

Total hours: 30

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	An ability to function effectively in teams to accomplish a common goal.
CO2	An ability to apply knowledge of computing and engineering to evaluate project requirements.
CO3	An ability to design, implement and evaluate a computer-based system, process, component, or program to meet desired needs.
CO4	An ability to analyze the local and global impact of computing on individuals, organizations, and society.
CO5	Write technical report and deliver presentation by applying different visualization tools and evaluation metrics.
CO6	An ability to communicate effectively with end user.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	3	2	-	1	2	-
CO2	3	-	1	2	-	-	-	-	-	-	-	-	3	-
CO3	-	2	3	-	1	-	-	-	-	-	-	-	2	-
CO4	-	-	-	-	-	3	2	-	-	1	-	-	2	-
CO5	-	-	-	-	-	-	1	-	-	2	-	3	-	2
CO6	-	-	-	-	-	-	3	1	-	-	-	2	-	3

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

B. Instructional Method and Pedagogy:

- Students will select related topic based on subjects they learnt and other literatures like books, periodicals, journals and various internet resources.
- Students can select the topic based on the research areas of available supervisor.
- Each student has to prepare a synopsis, SRS, SPMP, Design documents and Final Report in prescribed format only. The report typed on A4 sized sheets and bound should be submitted after approval by the supervisor/guide and endorsement of the Head of Department.
- Performance of students at the Project will be assessed by the department faculty. The supervisor/guide based on the quality of work and preparation and understanding of the candidate shall do an assessment of the project.

C. Recommended Study Material:

1. Reading Materials, web materials, Project reports with full citations
2. Books, magazines & Journals of related topics
3. Various software tools and programming languages compiler related to topic.

❖ Web Materials:

1. www.ieeexplore.ieee.org
2. www.sciencedirect.com
3. www.elsevier.com
4. <http://spie.org/x576.xm>

CS343: SUMMER INTERNSHIP-I

Credits and Hours:

Teaching Scheme	Theory	Practical	Project	Total	Credit
Hours/week	0	0	-	90	3
Marks	0	0	-	150	

A. Instruction Method and Pedagogy:

- Summer internship shall be at least 90 hours during the summer vacation only.
- Department/Institute will help students to find an appropriate company/industry/organization for the summer internship.
- The student must fill up and get approved a Summer Internship Acceptance form by the company and provide it to the Coordinator of the department within the specified deadline.
- Students shall commence the internship after the approval of the department Coordinator. Summer internships in research centers is also allowed.
- During the entire period of internship, the student shall obey the rules and regulations of the company/industry/organization and also those of the University.
- Due to inevitable reasons, if the student will not able to attend the internship for few days with the permission of the supervisor, the department Coordinator should be informed via e-mail and these days should be compensated later.
- The student shall submit two documents to the Coordinator for the evaluation of the summer internship:
 - Summer Internship Report
 - Summer Internship Assessment Form
- Upon the completion of summer internship, a hard copy of “Summer Internship Report” must be submitted to the Coordinator by the first day of the new term.
- The report must outline the experience and observations gained through practical internship, in accordance with the required content and the format described in this guideline. Each report will be evaluated by a faculty member of the department on a satisfactory/unsatisfactory basis at the beginning of the semester.
- If the evaluation of the report is unsatisfactory, it shall be returned to the student for revision and/or rewriting. If the revised report is still unsatisfactory the student shall be requested to repeat the summer internship.

B. Format of Summer Internship Report

The report shall comply with the summer internship program principles. Main headings are to be centered and written in capital boldface letters. Sub-titles shall be written in small letters and boldface. The typeface shall be Times New Roman font with 12pt. All the margins shall be 2.5cm. The report shall be submitted in printed form and filed. An electronic copy of the report shall be recorded in a CD and enclosed in the report. Each report shall be bound in a simple wire vinyl file and contain the following sections:

- Cover Page
- Page of Approval and Grading
- Abstract page: An abstract gives the essence of the report (usually less than one page). Abstract is written after the report is completed. It must contain the purpose and scope of internship, the actual work done in the plant, and conclusions arrived at.
- TABLE OF CONTENTS (with the corresponding page numbers)
- LIST OF FIGURES AND TABLES (with the corresponding page numbers)
- DESCRIPTION OF THE COMPANY: Summarize the work type, administrative structure, number of employees (how many engineers, under which division, etc.), etc. Provide information regarding
 - Location and spread of the company
 - Number of employees, engineers, technicians, administrators in the company
 - Divisions of the company
 - Your group and division
 - Administrative tree (if available)
 - Main functions of the company
 - Customer profile and market share
- INTRODUCTION: In this section, give the purpose of the summer internship, reasons for choosing the location and company, and general information regarding the nature of work you carried out.
- PROBLEM STATEMENT: What is the problem you are solving, and what are the reasons and causes of this problem.
- SOLUTION: In this section, describe what you did and what you observed during the summer internship. It is very important that majority of what you write should be based on what you did and observed that truly belongs to the company/industry/organization.

- **CONCLUSIONS:** In the last section, summarize the summer internship activities. Present your observations, contributions and intellectual benefits. If this is your second summer internship, compare the first and second summer internships and your preferences.
- **REFERENCES:** List any source you have used in the document including books, articles and web sites in a consistent format.
- **APPENDICES:** If you have supplementary material (not appropriate for the main body of the report), you can place them here. These could be schematics, algorithms, drawings, etc. If the document is a datasheet and it can be easily accessed from the internet, then you can refer to it with the appropriate internet link and document number. In this manner you don't have to print it and waste tons of paper.

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Ability to integrate existing and new technical knowledge for industrial application.
CO2	Executing work with team and teammates from other disciplines
CO3	Get practices and experience related to professional and ethical issues in the work environment
CO4	Experience of demonstrating the impact of the internship on their learning and professional development.
CO5	Understanding of lifelong learning processes through critical reflection of internship experiences.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	1	-	-	1	-	-	-	-	-	-	-	-	-
CO2	-	-	-	-	-	-	-	-	3	2	1	-	-	2
CO3	-	-	-	-	-	-	1	3	-	1	-	-	-	1
CO4	-	-	-	-	-		3	1	-	-	-	-	1	-
CO5	-		-	-	-	1	-	-	-	-	-	3	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) If

there is no correlation, put “-”

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HS 124.01 A: PROFESSIONAL COMMUNICATION (HSS ELECTIVE-III)

A. Credits and Schemes

Se m.	Course Code	Course Name	Cre d its	Teaching Scheme	Evaluation Scheme				
				Contact Hours/W eek	Theory		Practical		Tot al
					Inter nal	Exter nal	Inter nal	Exter nal	
V	HS124. 01 A	Professional Communica tion	02	02	--	--	30	70	100

B. Course Outline

Module No.	Title/Topic	Classroom Contact Sessions
1	An Introduction Professional Communication • Concept & Applications of Professional Communication • Principles of Professional Communication • Communication networks: personal sanctum, professional sanctum, inner circle, and outer circle; managing the networks • Communication strategies: communicator, audience, message, channel choice, culture	04
2	Professional Communication and Rhetorics • Concepts of “Communication” and “Professional Communication” and “Rhetorics” • Orientation towards the Concepts of Professional Communication and Rhetorics (Speaking) • Importance of Ethos, Logos, and Pathos in Professional Communication • Principles of Professional Communication (visual, oral and non-verbal)	04

3	Cross-cultural Communication and Globalization • Basic Concepts: Culture, Globalization and Cross-cultural Communication • Communicating with People of Different Cultures • Conflicts in Cross-cultural Communication and Tactics / techniques to resolve them	04
4	Written Professional Communication • Importance of Written Professional Communication • Letter Writing, E-mail Writing, Report Writing • Resume Building	06
5	Academic Writing • Importance of Academic Writing • Research Paper Writing • Article/ Review Writing • Reference and Citation	06
6	Effective Presentation Strategies • Why and How in Presentation • Audience Analysis and Supporting Material • Presentation Mechanics and Presentation Process • Managing Yourself during Q and A Session • Fundamentals of Persuasions	06
Total		30

C. Pedagogy

Teaching will be facilitated by reading material, discussions, task-based learning, projects, assignments and interpersonal activities like group work, independent and collaborative study projects and presentations, etc.

D. Evaluation

Internal Evaluation

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Assignment	02	05	10
2	Project	01	15	15
3	Attendance			05
Total				30

External Evaluation

University Practical Examination will be for 70 marks to be conducted at the end of the semester. Details are:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Practical / Viva	01	70	70
Total				70

Course Outcome (COs):

After completion of the course, the student would :

CO1	Gain basic conceptual understanding of communication skills in Professional settings
CO2	Develop awareness and competence in communicating across cultures in professional settings
CO3	Develop confidence and competence in speaking in formal interviews and to make formal presentations
CO4	Develop necessary soft skills to work collaboratively in a group and participate in a group Discussion for employment and in the workplace.
CO5	Develop team and group dynamics to work well in multi-disciplinary and cross-cultural work environment
CO6	Develop writing skills to prepare CV's, Resumes, Statement of Purpose and preparation of other formal documents.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO2	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO3	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO4	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO5	-	-	-	-	-	-	-	-	2	3	-	-	-	2
CO6	-	-	-	-	-	-	-	-	-	3	-	-	-	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

E. Reference Books

- Eckhouse, B. E. (1999). *Competitive Communication: A Rhetoric for Modern Business*. Oxford University Press. (for Module - II)
- Koneru, A. (2008). *Professional Communication*. New Delhi: Tata Mcgraw-Hill.(for Module I & III)
- Meenakshi Raman, P. S. (2006). *Business Communication*. Meenakshi Raman, Prakash Singh. (for module I and IV)
- Parul Popat, K. K. (2015). *Communication Skills*. New Delhi: Pearson. (for Module V & VI)
- Sanjay Kumar, P. L. (2015). *Communication Skills*. Oxford University Press India.(for Module I, III and V)

B. Tech. (Computer Science & Engineering) Programme

SYLLABI **(Semester – 6)**

CS344: MACHINE LEARNING

A. Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	4
Marks	100	50	-	150	

A. Outline of the Course:

Sr. No.	Title of the unit	Minimum number of Hours
1.	Fundamental concepts and Statistical Learning Techniques	06
2.	Neural Networks	08
3.	Bayesian Learning	06
4.	Supervised and Unsupervised Learning	10
5.	Reinforcement learning	02
6.	Kernel Methods	03
7.	Deep Neural Networks	10

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

1	Fundamental concepts and Statistical Learning Techniques	06 hours	10%
	Fundamental concepts: Introduction to Data science, Theory and practices in machine learning, Designing a Learning System, Issues in Machine Learning, Applications of ML, Global Developments of ML, Key challenges to adoption of ML in India.		
	Statistical Learning Techniques: Descriptive statistics, Simple Linear Regression, ANOVA, Logistic Regression, Multi Linear regression, Correlation, Moving Average, Random Number Generation, Histogram Smoothing, Sampling, Regularization, Rank Percentile.		
2	Neural Networks	08 hours	20%
	Neurons and biological motivation. Linear threshold units. Perceptrons: representational limitation and gradient descent training, Perceptron learning rule, Hebbian learning rule, Delta Learning rule, Multilayer networks and Backpropagation Learning Algorithm, Feed Forward, Activation Functioning, Types of Neural Network Architecture.		
3	Bayesian Learning	06 hours	15%
	Bayes Theorem, Bayes Theorem and Concept Learning, Maximum Likelihood and Least Squared Error Hypothesis, Maximum likelihood hypothesis for Predicting probabilities, Minimum Description Length Principle, Bayes Optimal Classifier, Gibbs Algorithm, Naïve Bayes Classifier, Bayesian Belief Network, EM Algorithm.		
4	Supervised and Unsupervised Learning	10 hours	20%
	Supervised Learning: Classification, Decision Tree, Naïve Bias, Support Vector Machine, Neural Network, K- Nearest Neighbour.		
	Unsupervised Learning: Clustering, Nonparametric Methods, K-means, Hierarchical clustering, Density based clustering		
5	Reinforcement learning	02 hours	05%
	Q Learning, Non deterministic rewards and Actions.		
6	Kernel Methods	03 hours	10%
	Support Vector Machine, Sparse kernel machines, Bias-Variance trade-off. Regularization and model/feature selection, Sampling Methods.		
7	Deep Neural Networks	10 hours	20%

Introduction to Deep Learning, Deep Neural Network, Restricted Boltzmann machine, Convolution Neural Network, AutoEncoders, Deep Belief Network, Recurrent Neural Network, Transfer learning.

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Apply basic concepts of Machine Learning and Understanding of standard learning algorithms.
CO2	Analyse mathematical modelling of various Machine Learning algorithms.
CO3	Understanding challenges of machine learning like data characteristics, model selection, and model complexity.
CO4	Identify strengths and weaknesses of machine learning techniques suitable for a given problem domain and data set.
CO5	Design and implement of various machine learning algorithms in a range of real-world applications.
CO6	Evaluate and interpret the results of learning algorithms.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	-	-	-	-	-	-	-	-	-	3	-
CO2		2	-	-	-	-	-	-	-	-	-	-	3	-
CO3	3	-	-	-	-	-	3	-	-	-	-	-	-	-
CO4	3	2	-	-	-	-	-	-	-	-	-	-	2	-
CO5	3	-	2	2	2	-	-	-	-	-	2	-	3	-
CO6	2	-	-	-	-	-	-	-	-	-	-	-	3	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

C. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Minimum 5 experiments shall be there in the laboratory related to course contents.
- Research / technical papers in relevant areas must be covered.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

D. Recommended Study Material:

❖ Text Books:

1. Machine Learning, Tom Mitchell, McGraw Hill, 1997. ISBN 0070428077
2. Ethem Alpaydin, "Introduction to Machine Learning", MIT Press, 2004

❖ Reference Books:

1. Christopher M. Bishop, "Pattern Recognition and Machine Learning", Springer, 2006.
2. Richard O. Duda, Peter E. Hart & David G. Stork, "Pattern Classification. Second Edition", Wiley & Sons, 2001.
3. Trevor Hastie, Robert Tibshirani and Jerome Friedman, "The elements of statistical learning", Springer, 2001.
4. Richard S. Sutton and Andrew G. Barto, "Reinforcement learning: An introduction", MIT Press, 1998.

CS345: CRYPTOGRAPHY & NETWORK SECURITY

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	4	2	-	6	5
Marks	100	50	-	150	

A. Outline of the Course:

Sr No.	Title of the unit	Minimum number of hours
1.	Classical Cryptography	04
2.	Shannon's Theory	03
3.	Block Ciphers And The Advanced Encryption Standard	05
4.	Cryptographic Hash Functions	05
5.	The RSA Cryptosystem And Factoring Integers	05
6.	Public-Key Cryptography And Discrete Logarithms	05
7.	Signature Schemes	05
8.	Pseudo-Random Number Generation	03
9.	Key Distribution & Key Agreement Schemes	05
10.	Public-Key Infrastructure & Multicast Security	05

Total hours (Theory): 60

Total hours (Lab): 30

Total hours: 90

B. Detailed Syllabus:

- | | | |
|--|-----------------|-------------|
| 1. Classical Cryptography | 04 hours | 09 % |
| 1.1 Introduction: Some Simple Cryptosystems | | |
| 1.2 Cryptanalysis | | |
| 2. Shannon's Theory | 03 hours | 07 % |
| 2.1 Introduction | | |
| 2.2 Elementary Probability Theory | | |
| 2.3 Perfect Secrecy | | |
| 2.4 Entropy | | |
| 2.5 Product Cryptosystems | | |
| 3. Block Ciphers And The Advanced Encryption Standard | 05 hours | 11% |
| 3.1 Introduction | | |
| 3.2 Substitution-Permutation Networks | | |
| 3.3 The Data Encryption Standard | | |
| 3.4 The Advanced Encryption Standard | | |
| 3.5 Modes of Operation | | |
| 4. Cryptographic Hash Functions | 05 hours | 11% |
| 4.1 Hash Functions and Data Integrity | | |
| 4.2 Security of Hash Functions | | |
| 4.3 Iterated Hash Functions | | |
| 4.4 Message Authentication Codes | | |
| 5. The RSA Cryptosystem And Factoring Integers | 05 hours | 11% |
| 5.1 Introduction to Public-key Cryptography | | |
| 5.2 The RSA Cryptosystem | | |
| 5.3 Primality Testing | | |
| 5.4 Factoring Algorithms | | |
| 5.5 Other Attacks on RSA | | |
| 6. Public-Key Cryptography And Discrete Logarithms | 05 hours | 11% |
| 6.1 The ElGamal Cryptosystem | | |
| 6.2 Algorithms for the Discrete Logarithm Problem | | |
| 6.3 Lower Bounds on the Complexity of Generic Algorithms | | |
| 6.4 Finite Fields | | |
| 6.5 Elliptic Curves | | |

7. Signature Schemes **05 hours 11%**

- 7.1 Introduction
- 7.2 Security Requirements for Signature Schemes
- 7.3 The ElGamal Signature Scheme
- 7.4 Variants of the ElGamal Signature Scheme

8. Pseudo-Random Number Generation **03 hours 07%**

- 8.1 Introduction and Examples
- 8.2 The Blum-Blum-Shub Generator
- 8.3 Probabilistic Encryption

9. Key Distribution & Key Agreement Schemes **05 hours 11%**

- 9.1 Introduction
- 9.2 Diffie-Hellman Key Predistribution
- 9.3 Key Distribution Patterns
- 9.4 Session Key Distribution Schemes
- 9.5 Diffie-Hellman Key Agreement
- 9.6 Key Agreement Using Self-Certifying Keys
- 9.7 Encrypted Key Exchange

10. Public-Key Infrastructure&Multicast Security **05 hours 11%**

- 10.1 Introduction: What is a PKI?
- 10.2 Certificates
- 10.3 The Future of PKI?
- 10.4 Identity-Based Cryptography
- 10.5 Introduction to Multicast Security
- 10.6 Broadcast Encryption
- 10.7 Multicast Re-Keying

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Define various security goal and understand the security policies such as the CIA triad of Confidentiality, Integrity and Availability.
CO2	Classify various forms of security attacks, where they arise, and appropriate tools or mechanism to quantify them.
CO3	Illustrate a basic understanding of cryptography, how it has evolved, and evaluate symmetric key encryption techniques used today.
CO4	Distinguish modern symmetric encryption standard, key distribution scenario and analyse effectiveness in todays' environment.

CO5	Evaluate Asymmetric key encryption techniques, key distribution scenario and calculate public and private components of asymmetric key encryption techniques.
CO6	Develop message integrity and message authentication of message digest.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	2	-	-	3	1	-	2	1	-	-	-	-	-
CO2	2	3	3	2	3	1	-	-	-	-	-	-	1	-
CO3	2	2	3	2	3	2	-	-	2	1	-	-	2	-
CO4	2	2	3	2	3	-	2	-	2	1	-	-	3	1
CO5	3	2	3	2	1	-	2	1	1	-	-	-	3	1
CO6	2	2	3	1	-	-	2	2	1	-	2	-	2	2

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) If

there is no correlation, put “-”

C. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries 5 Marks weightage.
- Two internal exams will be conducted and average of the same will be converted to equivalent of 15 Marks as a part of internal theory evaluation.

- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval. It carries a weightage of 5 Marks as a part of internal theory evaluation.
- Surprise tests/Quizzes/Seminar will be conducted which carries 5 Marks as a part of internal theory evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.

D. Recommended Study Material:

❖ Text Books:

1. Douglas R. Stinson, Cryptography: Theory and Practice, Chapman & Hall/CRC

❖ Reference Books:

1. William Stallings, Cryptography And Network Principles And Practice, Prentice Hall, Pearson Education Asia
2. AtulKahate, Cryptography & Network Security, The McGraw-Hill Companies
3. Behrouz A. Forouzan, Cryptography and Network Security, McGraw-Hill Companies

❖ Reference Links/ e-content:

1. <http://people.csail.mit.edu/rivest/crypto-security.html>
2. <http://www.cryptix.org/>
3. <http://www.cryptocd.org/>
4. <http://www.cryptopp.com/>

CE349: THEORY OF COMPUTATION

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	0	-	3	3
Marks	100	0	-	100	

A. Outline of the Course:

Sr. No.	Title of the unit	Minimum Number of Hours
1	Introduction	02
2	Mathematical Terms and Theory	08
3	Regular Grammar & Languages, Regular Expression, Finite Automata	22
4	Context Free Grammar & Languages, Push down Automata	18
5	Turing Machine, Recursively Enumerable Languages	08
6	Tractable and Intractable Problems	02

Total Hours (Theory): 45

Total Hours (Lab): 00

Total Hours: 45

B. Detailed Syllabus:

- | | | |
|---|-----------------|-------------|
| 1. Introduction | 02 Hours | 03 % |
| Alphabet, String, Language, Formal Grammar, Chomsky Hierarchy, Introduction to Automata | | |
| 2. Mathematical Terms and Theory | 08 Hours | 14 % |
| Basic Mathematical Notations, Set Theory, Logical Statement, Function, Proof, Relation, Closure of Relation, Mathematical Inductions, Recursive Definitions, | | |
| 3. Regular Grammar & Languages, Regular Expression, Finite Automata | 22 Hours | 36 % |
| Regular Language, Regular Expressions, Applications, Chomsky Hierarchy, Finite Automata, Nondeterministic Finite Automata Kleen's Theorem, Automata with Output (Moore Machine, Mealy Machine), Properties of Regular Languages (Pumping Lemma Closure Property, Decision Algorithm, The myhill-nerode Theorem) | | |
| 4. Context Free Grammar & Languages, Push down Automata | 18 Hours | 30 % |
| The Chomsky, Notion of Grammars and Languages Generated by Grammars, CFG, CFL, Regular Language and Regular Grammar Derivation Tree and Ambiguity, BNF, CNF, GNF, CFL properties (Pumping Lemma, Closure Property, Decision Algorithm Intersections and Complements of CFL, Non-CFL, Definition, DPDA, NPDA, Equivalence of CFG and PDA | | |
| 5. Turing Machine, Recursively Enumerable Languages | 08 Hours | 14 % |
| Definition, Model of Computation, Combining TM, Variations of TM, Non Deterministic TM, Universal TM, Recursively Enumerable and Recursive, Enumerable Languages, Context sensitive languages | | |
| 6. Tractable and Intractable Problems | 02 Hours | 03 % |
| Growth Rate and Functions, Time and Space Complexity, | | |

Complexity Classes, Tractable and Possibly Intractable Problems,
P and NP Completeness

C. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Two internal exams will be conducted and average of the same will be converted to equivalent of 15 Marks as a part of internal theory evaluation.
- Assignments/Surprise tests/Quizzes/Seminar will be conducted which carries 5 Marks as a part of internal theory evaluation.

D. Student Learning Outcome:

After completion of the course students will be able to

- Understand the basic concepts and application of Theory of Computation.
- Apply knowledge in the computer field to solve computational problems and in the field of Natural Language Processing.

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Apply basic concepts of theory of computation in the computer field in order to solve computational problems.
CO2	Construct algorithms for different problems and argue formally about correctness on different restricted machine models of computation.
CO3	Analyze and design finite automata, pushdown automata and Turing machine for formal languages.
CO4	Apply rigorously formal mathematical methods to prove properties of languages, grammars and automata.
CO5	Identify limitations of some computational models and possible solutions.
CO6	Design context free grammars for formal languages.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	1	-	-	-	-	-	-	-	-	-	-	3	-
CO2	-	-	3	-	-	-	-	-	-	-	-	-	2	-
CO3	3	2	2	-	-	-	-	-	-	-	-	-	3	-
CO4	2	2	2	-	-	-	-	-	-	-	-	-	1	-
CO5	2	1	1	-	-	-	-	-	-	-	-	-	-	-
CO6	2	-	-	-	-	-	-	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) If there is no correlation, put “-”

E. Recommended Study Material:

❖ Text Books:

1. Introduction to Languages and Theory of Computation, John C. Martin, TMH

❖ Reference Books:

1. An introduction to automata theory and formal languages, Adesh K. Pandey, S.K. Kataria & Sons
2. Introduction to computer theory, Deniel I. Cohen, John Wiley & Sons Inc
3. Computation: Finite and Infinite, Marvin L. Minsky, Prentice-Hall
4. “An introduction to Formal Languages and Automata”, Peter Linz, 6th edition, Jones & Bartlett Learning
5. “Introduction to the Theory of Computation”, Michael Sipser, 3rd edition, Cengage Learning.

❖ **Web Materials:**

1. http://en.wikipedia.org/wiki/Theory_of_computation
2. https://www.youtube.com/playlist?list=PLEbnTDJUr_IdM____FmDFBz0zCsOFxfK
3. <http://nptel.ac.in/courses/106103070/>
4. <http://nptel.ac.in/courses/106104028/>
5. <http://nptel.ac.in/courses/106106049/>
6. <https://www.youtube.com/watch?v=4GLC-s0PQLY>

CS346: SOFTWARE ENGINEERING

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the unit	Minimum Number of Hours
1	Introduction to Software and Software Engineering	4
2	Agile Development	4
3	Managing Software Project	5
4	Requirement Analysis and Specification	4
5	Software Design	5
6	Software Coding & Testing	6
7	Quality Assurance and Management	5
8	Software Maintenance and Configuration Management	5
9	Introduction to SaaS	3
10	Advanced Topics in Software Engineering	4

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

1	Introduction to Software and Software Engineering	04 Hours	9%
1.1	The Evolving Role of Software		
1.2	Software: A Crisis on the Horizon and Software Myths		
1.3	Software Engineering: A Layered Technology		
1.4	Software Process Models, The Linear Sequential Model, The Prototyping Model, The RAD Model, Evolutionary Process Models, Agile Process Model		
1.5	Component-Based Development, Process, Product and Process		
2.	Agile Development	04 Hours	9%
2.1	Agility and Agile Process model		
2.2	Extreme Programming		
2.3	Other process models of Agile Development and Tools		
3	Managing Software Project	05 Hours	11%
3.1	Software Metrics (Process, Product and Project Metrics)		
3.2	Software Project Estimations		
3.3	Software Project Planning (MS Project Tool)		
3.4	Project Scheduling & Tracking		
3.5	Risk Analysis & Management(Risk Identification, Risk Projection, Risk Refinement ,Risk Mitigation)		
4	Requirement Analysis and Specification	04 Hours	9%
4.1	Understanding the Requirement		
4.2	Requirement Modeling		
4.3	Requirement Specification (SRS)		
4.4	Requirement Analysis and Requirement Elicitation		
4.5	Requirement Engineering		
5	Software Design	05 Hours	11%
5.1	Design Concepts and Design Principal		
5.2	Architectural Design		
5.3	Process and Component Level Design (Function Oriented Design, Object Oriented Design-UML)		

	(MS Visio Tool)		
5.4	User Interface Design		
5.5	Web Application Design, Design Patterns		
6.	Software Coding & Testing	06 Hours	13%
6.1	Coding Standard and Coding Guidelines		
6.2	Code Review		
6.3	Software Documentation		
6.4	Testing Strategies		
6.5	Testing Techniques and Test Case, Test Suites Design		
6.6	Testing Conventional Applications		
6.7	Testing Object-Oriented Applications		
6.8	Testing Web and Mobile Applications, Testing Tools (Win runner, Load runner)		
7	Quality Assurance and Management	05 Hours	11%
7.1	Quality Concepts and Software Quality Assurance		
7.2	Software Reviews (Formal Technical Reviews)		
7.3	Software Reliability		
7.4	The Quality Standards: ISO 9000, CMM, Six Sigma for SE.		
7.5	SQA Plan		
8	Software Maintenance and Configuration Management	05 Hours	11%
8.1	Types of Software Maintenance, Re-Engineering, Reverse Engineering, Forward Engineering		
8.1	The SCM Process, Identification of Objects in the Software Configuration		
8.2	Version Control and Change Control		
9.	Introduction to SaaS and SOA	03 Hours	7%
9.1	Product Lifetime: Independent Product Vs. Continues Improvement		
9.2	Software as a Service		
9.3	Service Oriented Architecture		
9.4	Cloud Computing		
9.5	SaaS Architecture		
10	Advanced Topics in Software Engineering	04 Hours	9%

- 10.1 Component-Based Software Engineering, Client/Server Software Engineering, Web Engineering, Reengineering, Computer-Aided Software Engineering, Real-Time Software Engineering
- 10.2 Software Process Improvement
- 10.3 Design Thinking

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand basics about software engineering principles, methods and practices and to analyze software requirement specification Prepare, SRS (Software Requirement Specification) document and SPMP (Software Project Management Plan) document.
CO2	Apply the concept of Functional Oriented and Object Oriented Approach for Software Design, To explain the software design strategies and to apply software measurement and metrics using Function point, Cyclomatic complexity and Heilstead software science measures.
CO3	Recognize how to ensure the quality of software product, different quality standards and software review techniques.
CO4	Formulate problem by following Software Testing Life Cycle. Apply various testing techniques and test plan in. Design Manual Test cases for Software Project. Use automation testing tool students will be able test the software.
CO5	Able to understand modern Agile Development and Service Oriented Architecture Concept of Industry.
CO6	Analyze software risk with estimation parameters such as cost, effort, schedule/duration and understand the concepts of software maintenance, reengineering, reverse engineering, software configuration management.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	-	2	1	-	-	3	-	-	-	-	2	-
CO2	2	2	2	1	2	1	-	-	2	-	1	-	2	-
CO3	1	2	3	2	1	1	1	-	2	1	1	1	-	-
CO4	1	3	2	-	1	-	-	-	1	-	-	1	2	-

CO5	-	-	-	-	-	1	-	2	1	1	1	1	1	1
CO6	-	2	-	1	-	-	1	-	2	2	1	2	1	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) If there is no correlation, put “-”

C. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries 5 Marks weightage.
- Two internal exams will be conducted and average of the same will be converted to equivalent of 15 Marks as a part of internal theory evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval. It carries a weightage of 5 Marks as a part of internal theory evaluation.
- Surprise tests/Quizzes/Seminar will be conducted which carries 5 Marks as a part of internal theory evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.
- Students have to undergone MOOCS Classes by Armando Fox and David Patterson

D. Recommended Study Material:

❖ Text Books:

1. Roger S. Pressman, Software engineering- A practitioner's Approach, McGraw-Hill International Editions
2. Ivar Jacobson, Object Oriented Software Engineering: A Use Case Driven Approach, 1st edition

❖ Reference Books:

1. Engineering Software as a Service An Agile Software Approach, Armando Fox and David Patterson
2. Ian Sommerville, Software engineering, Pearson education Asia
3. Pankaj Jalote, An Integrated Approach to Software Engineering by, Springer
4. Rajib Mall, Fundamentals of software Engineering, Prentice Hall of India.
5. John M Nicolas, Project Management for Business, Engineering and Technology, Elsevier

❖ Web Materials:

1. www.en.wikipedia.org/wiki/Software_engineering
2. www.win.tue.nl
3. www.rspa.com/spi
4. www.onesmartclick.com/engsineering/software-engineering.html
5. www.sei.cmu.edu
6. <https://www.edx.org/school/uc-berkeleyx>

CS373: INTERNET OF THINGS (PE-I1)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	4
Marks	100	50	-	150	

A. Outline of the Course:

Sr No.	Title of the unit	Minimum number of Hours
1.	Introduction to Internet of Things	10
2.	Prototyping IoT Devices	06
3.	IOT & M2M	04
4.	Domain Specific IoTs	10
5.	Internet of Things Privacy, Security and Governance	07
6.	IoT Physical Servers & Cloud Offerings	08

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 7

B. Detailed Syllabus:

- | | | | |
|---|---|-----------------|------------|
| 1 | Introduction to Internet of Things | 10 hours | 20% |
| | Definition & Characteristics of IoT, Physical Design of IoT Logical Design of IoT, IoT Enabling Technologies, IoT Levels & Deployment Templates | | |
| 2 | Prototyping IoT Devices | 06 hours | 20% |
| | Sensors & Actuators, Microcontrollers, Arduino and Raspberry Pi | | |
| 3 | IoT & M2M | 04 hours | 10% |
| | M2M, Difference between IoT and M2M, SDN and NFV for IoT | | |
| 4 | Domain Specific IoTs | 10 hours | 20% |

Home Automation, Cities, Environment, Energy, Retail, Logistics, Agriculture, Industry, Health & Lifestyle

5 Internet of Things Privacy, Security and Governance

07 15%
hours

Introduction, Overview of Governance, Privacy and Security Issues, Contribution from FP7 Projects, Security, Privacy and Trust in IoT-Data-Platforms for Smart Cities, First Steps Towards a Secure Platform, Smartie Approach, Data Aggregation for the IoT in Smart Cities, Security

6 IoT Physical Servers & Cloud Offerings

08 15%
hours

Introduction to Cloud Storage Models & Communication APIs, WAMP - AutoBahn for IoT. Xively Cloud for IoT, Designing a RESTful Web API, SkyNet IoT Messaging Platform

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Interpret the vision of IoT from current business needs.
CO2	Compare and contrast the use of devices, gateways and data management in IoT.
CO3	Demonstrate a critical understanding of key technologies in sensing, data transmission and data processing for IoT.
CO4	Examine of the technologies and the standards relating to the Internet of Things and analyze basic protocols in sensor network
CO5	Evaluation of privacy management, challenges in designing and securing an IoT system.
CO6	Illustrate the application of IoT in industrial automation and identify real world design Constraints.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	2	2	-	-	-	-	-	-	-	-	-	-	-
CO2	3	2	3	2	1	-	-	-	-	-	-	-	2	-
CO3	1	2	3	-	1	-	-	-	-	-	-	-	2	-
CO4	2	1	2	1	2	-	-	-	-	-	-	-	3	-
CO5	-	1	2	1	1	-	-	-	-	-	-	-	2	-
CO6	2	1	3	-	1	-	-	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

C. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.

D. Recommended Study Material:

❖ Text Books:

1. “Internet of Things: Converging Technologies for Smart Environments and Integrated Ecosystems”, Ovidiu Vermesan, Peter Friess, River Publishers

❖ Reference Books:

1. Internet of Things: A hands on approach by Arhdeep Bahga and Vijay Madisetti.
2. Research papers from IEEE, Springer etc.
3. The Internet of Things-ITU.

❖ Web Materials:

1. <http://www.vs.inf.ethz.ch/res/show.html?what=iot> – For Research Papers
2. www.ieee.org – For standards and technical research papers

CS375: SERVICE ORIENTED COMPUTING (PE-II)

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Overview of SOA	03
2.	SOA — Architecture Fundamentals	06
3.	Web Services and Contemporary SOA (Part I: Activity Management and Composition)	06
4.	Web Services and Contemporary SOA (Part II: Advanced Messaging, Metadata, and Security)	08
5.	Building SOA (Planning and Analysis)	06
6.	Building SOA (Technology and Design)	10
7.	Fundamental WS-* Extensions	06

Total hours (Theory): 45Hrs.

Total hours (Lab): 30Hrs.

Total hours: 75Hrs.

B. Detailed Syllabus:

1. Overview of SOA	03	07
	hours	%
1.1 The Promise of SOA		
1.2 The Challenges of SOA: Reuse, Efficiency in Development, Integration of Applications and Data, Agility, Flexibility, and Alignment.		
1.3 Meeting the Challenge Reference Architecture Common Semantics Governance Business Process Modeling Design-Time Service Discovery Model-Based Development		
2. SOA — Architecture Fundamentals	06	13
	hours	%
2.1 Architectural Styles, Architectural Principles and Practices		
2.2 SOA and Other Architectures, Enterprise Architecture, Software Architecture, EA, 4+1, and Services		
2.3 What Is a Service?, Service Characteristics: Service, Granularity, Service Dimensions.		
2.4 Loose Coupling Is King: Location Transparency, Interface and Implementation, Data, Versioning, Interoperability and Platform Independence, Usage, Assumptions, and Knowledge, Common Service Patterns, Service Types and Purpose, SOA Reference Architecture		
3. Web Services and Contemporary SOA (Part I: Activity Management and Composition)	06	13%
	hours	
3.1 Message exchange patterns		
3.2 Service activity		
3.3 Coordination		
3.4 Atomic transactions		
3.5 Business activities		
3.6 Orchestration		
4. Web Services and Contemporary SOA (Part II: Advanced Messaging, Metadata, and Security)	08	19%
	hours	

4.1	Addressing		
4.2	Reliable messaging		
4.3	Correlation		
4.4	Policies		
4.5	Metadata exchange		
4.6	Security		
4.7	Notification and eventing		
5.	Building SOA (Planning and Analysis)	06	13%
		hours	
5.1	SOA Delivery Strategies: SOA delivery lifecycle phases, The top-down strategy, The bottom-up strategy		
5.2	Service Modeling: Service modeling guidelines, Classifying service model logic		
6.	Building SOA (Technology and Design)	10	22%
		hours	
6.1	WSDL-related XML Schema language basics, WSDL language basics, SOAP language basics		
6.2	Composition Guidelines: Steps to composing SOA, Considerations for choosing service layers, Considerations for positioning core SOA standards		
6.3	Service Design: Service design overview, Entity-centric business service design (a step-by-step process), Application service design, Task-centric business service design		
6.4	Business Process Design: WS-BPEL language basics, WS-Coordination overview		
7.	Fundamental WS-* Extensions	06	13%
		hours	
7.1	WS-Addressing, WS-ReliableMessaging, WS-Policy, WS-MetadataExchange, WS-Security		
7.2	SOA Platforms: SOA support in J2EE, SOA support in .NET		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Define various promise and challenges of service oriented architecture.
CO2	Classify various forms of architectures, integration and interoperability of architecture principles.
CO3	Illustrate a basic understanding of activity management, how it has evolved, and evaluate composition and message exchanging.
CO4	Classifying delivery life cycle phase and analyse service model logic.
CO5	Evaluate web services using SOAP, WSDL and create application service design.
CO6	Explore platforms such as J2EE and .Net for reliable messaging, security and policies for efficient service discovery.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	2	-	-	3	1	-	2	1	-	-	-	-	-
CO2	2	3	3	2	3	1	-	-	-	-	-	-	1	-
CO3	2	2	3	2	3	2	-	-	2	1	-	-	2	-
CO4	2	2	3	2	3	-	2	-	2	1	-	-	3	1
CO5	3	2	3	2	1	-	2	1	1	-	-	-	3	1
CO6	2	2	3	1	-	-	2	2	1	-	2	-	2	2

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

C. Instructional Method and Pedagogy:

- Lectures will be taken in class room with the use of multi-media presentations and black board – mix of both.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval. It carries a weightage of 5 Marks as a part of internal theory evaluation.
- Surprise tests/Quizzes/Seminar will be conducted which carries 5 Marks as a part of internal theory evaluation.
- The course includes a laboratory, where students have an opportunity to build an

appreciation for the concepts being taught in lectures.

- Experiments/Tutorials related to course content will be carried out in the laboratory.

D. Recommended Study Material:

❖ Text Books:

1. Mike Rosen, Boris Lublinsky, Kevin T. Smith, Marc J. Balcer "Applied SOA:Service-Oriented Architectureand Design Strategies" Wiley Publishing, Inc, 2008
2. Thomas Erl, “Service-Oriented Architecture: Concepts, Technology, and Design”, Prentice Hall Publication, 2008

❖ Reference Books:

1. Thomas Erl, “Service-Oriented Architecture: A Field Guide to Integrating XML and Web Services”, Prentice Hall Publication, 2004
2. SanjivaWeerawarana, Francisco Curbera, Frank Leymann, Tony Storey, Donald F.Ferguson, “Web Services Platform Architecture: SOAP, WSDL, WS-Policy, WS-Addressing, WS-BPEL, WS-Reliable Messaging, and More”, Prentice Hall Publication, 2005
3. Eric Newcomer, Greg Lomow, “Understanding SOA with Web Services”, Addison Wesley Publication, 2004

❖ Reference Links/ e-content:

1. <https://www.ibm.com/developerworks/library/ws-top10/>
2. <http://www.oracle.com/technetwork/articles/javase/soa-142870.html>
3. <https://docs.oracle.com/javaee/6/tutorial/doc/gijti.html>.

CS374: MODERN NETWORKS (PE-II)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	0	5	4
Marks	100	50	0	150	

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Elements of Modern Networking	07
2	Requirements and Technology	10
3	SDN: Background and Motivation	10
4	SDN Data Plane and OpenFlow	05
5	SDN Control Plane&SDN Application Plane	13

Total hours (Theory): 45

Total hours (Lab): 30

Total hours: 75

B. Detailed Syllabus:

- | | | |
|---|-------------|-----|
| 1. Elements of Modern Networking | 07
Hours | 16% |
| The Networking Ecosystem, Example Network Architectures, Ethernet, Wi-Fi, 4G/5G Cellular, Network Convergence, Unified Communications | | |
| 2. Requirements and Technology | 10
Hours | 22% |
| Types of Network and Internet Traffic, Demand: Big Data, Cloud Computing, and Mobile Traffic, Requirements: QoS and QoE, Routing, Congestion Control, SDN and NFV, Modern Networking Elements | | |
| 3. SDN: Background and Motivation | 10
Hours | 22% |
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Evolving Network Requirements, The SDN Approach, SDN- and NFV-Related Standards

4. SDN Data Plane and OpenFlow 05
Hours 11%

SDN Data Plane, OpenFlow Logical Network Device, OpenFlow Protocol

5. SDN Control Plane&SDN Application Plane 13
Hours 29%

SDN Control Plane Architecture, ITU-T Model, OpenDaylight, REST, Cooperation and Coordination Among Controllers, SDN ApplicationPlane Architecture, Network Services Abstraction Layer, Traffic Engineering, Measurement and Monitoring, Security, Data Center Networking, Mobility and Wireless, Information-Centric Networking

• **Course Outcome (COs):**

At the end of the course, the students will be able to

CO1	Measure and Analyse different network parameters.
CO2	Understand working of application layer protocols.
CO3	Understand and Analyse transport layer services and its impacts on data rate.
CO4	Understand basic functionality of network layer devices.
CO5	Understand functionality of multiple access protocol.
CO6	Analyse traditional network and get familiar with Software defined networking.

• **Course Articulation Matrix:**

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	1	3	1	-	-	-	-	-	-	-	2	-
CO2	3	3	1	3	1	-	-	-	-	-	-	-	1	-
CO3	3	3	1	3	1	-	-	-	-	-	-	-	1	-
CO4	3	3	1	3	1	-	-	-	-	-	-	-	1	-
CO5	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO6	-	-	-	-	3	-	-	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) If there is no correlation, put “-”

C. Instructional Method and Pedagogy:

- Lectures will be taken in class room with the use of multi-media presentations and black board – mix of both.
- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Minimum 5 experiments shall be there in the laboratory related to course contents.
- Research / technical papers in relevant areas must be covered.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

D. Recommended Study Material:

❖ Text Books:

1. William Stallings, Florence Agboma, Sofiene Jelassi “Foundations of Modern Networking, SDN, NFV, QoE, IoT, and Cloud”; Pearson Publisher, ISBN-13:978-0-13-417539-3
2. Behrouz A. Forouzan, “TCP/IP Protocol Suite.”, Fourth Reprint, 2003; Tata McGraw Hill ISBN: 0-07-049551-3

❖ Reference Books:

1. Douglas E. Comer and David L. Stevens, “Internetworking with TCP/IP Volume- 2, Design, Implementation and Internals ”, Prentice Hall

❖ Web Materials:

1. <https://www.sdxcentral.com/>
2. <https://sdn.ieee.org/standardization>
3. <https://trac.ietf.org/trac/irtf/wiki/sdnrg>
4. <https://www.opennetworking.org/sdn-resources/openflow>
5. <https://www.opendaylight.org/>
6. <https://www.opennetworking.org/>

CS349: SOFTWARE GROUP PROJECT -IV

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	0	4	4	4
Marks	0	100	100	

A. Outline of the Course:

- Student at the beginning of a semester may be advised by his/her supervisor (s) for recommended courses.
- Students will work together in a team (at most **three**) with any programming language.
- Students are required to get approval of project definition from the department.
- After approval of project definition students are required to report their project work on weekly basis to the respective internal guide.
- Project will be evaluated at least once per week in laboratory Hours during the semester and final submission will be taken at the end of the semester as a part of continuous evaluation.
- Project work should include whole SDLC of development of software / hardware system as a solution of particular problem by applying principles of Software Engineering.
- Students have to submit project with following listed documents at the time of final submission.
 - a. Project Synopsis
 - b. Software Requirement Specification
 - c. SPMP
 - d. Final Project Report
 - e. Project Setup file with Source code
 - f. Project Presentation (PPT)
- A student has to produce some useful outcome by conducting experiments or project work.

Total Hours (Theory): 00

Total Hours (Lab): 60

Total Hours: 60

Course Outcome (COs):

At the end of the course, the students will be able to:

CO1	Identify and define the computing requirements of a problem to propose its appropriate solution.
CO2	Correlate knowledge of different subjects and apply it to implement solution of the problem.
CO3	Apply engineering and management principles to achieve project goal.
CO4	Prepare technical report and deliver presentation by applying different visualization tools and evaluation metrics.
CO5	Able to communicate effectively with a range of audiences.
CO6	Recognition of the need for and an ability to engage in continuing professional development.
CO7	Able to work and coordinate with different kind of people in the team

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	3	-	2	-	-	-	-	2	-	-	-	3	-
CO2	3	-	1	1	3	1	1	-	-	-	-	2	1	1
CO3	-	-	3	-	3	1	1	-	2	-	3	2	3	-
CO4	-	-	-	-	3	-	2	-	2	3	-	-	-	-
CO5	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO6	-	-	-	-	-	-	3	-	-	-	-	2	-	2
CO7	-	-	-	-	-	-	-	2	3	3	-	2	-	2

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) If there is no correlation, put “-”

B. Instructional Method and Pedagogy:

- Students will select related topic based on subjects they learnt and other literatures like books, periodicals, journals and various internet resources.
- Students can select the topic based on the research areas of available supervisor/guide.
- Each student has to prepare a synopsis, SRS, SPMP, Design documents and Final Report in prescribed format only. The report typed on A4 sized sheets and bound should be submitted after approval by the supervisor/guide and endorsement of the Head of Department.
- Performance of students at the Project will be assessed by the department faculty. The supervisor/guide based on the quality of work and preparation and understanding of the candidate shall do an assessment of the project.

C. Recommended Study Material:

1. Reading Materials, web materials, Project reports with full citations
2. Books, magazines & Journals of related topics
3. Various software tools and programming languages compiler related to topic

❖ Web Materials:

1. www.ieeexplore.ieee.org
2. www.sciencedirect.com
3. www.elsevier.com
4. <http://spie.org/x576.xml>

HS134 A: CONTRIBUTOR PERSONALITY DEVELOPMENT (HSS ELECTIVE-III)

A. Credits and Schemes:

Sem.	Course Code	Course Name	Credits	Teaching Scheme Contact Hours/Week	Evaluation Scheme				Total
					Theory		Practical		
					Internal	External	Internal	External	
VI	HS134 A	CONTRIBUTOR PERSONALITY DEVELOPMENT	02	02	--	--	30	70	100

B. Course Outline

Module No.	Title/Topic	Classroom Contact Hours
1	Concept of Personality: - Meaning of Personality - Types of Personality - Factors contributing to Personality - Personality Traits	06
2	Soft Skills and Personality Development: - Critical, Creative and Positive Thinking - Leadership, Assertiveness and Negotiation Skills - Self-Management - People's Skills - Building Relationship Skills - Being a Team Player	08
3	Developing Contributor Personality – Part I - Concept of Contributor - Characteristics of a Contributor - The Contributor's Vision of Success & Career - The Scope of Contribution - Embarking on the Journey to Contributor ship	06
4	Developing Contributor Personality – Part II - Focus on Values - Engage Deeply - Think in Enlightened Self-Interest - Practice Imaginative Sympathy - Demonstrate Trust Behavior - Developing a sense of duty and morality	06

5	Contemporary Issues in CPD - Contemporary Practices & Trends in Contributor Personality Development - Case Study & Presentations	04
	Total0	30

Course Outcome (COs):

After completion of the course, the student would :

CO1	Develop conceptual understanding of the term Personality and understand one's own personality Traits
CO2	Develop soft skills for holistic personality development for career and in personal life
CO3	Develop assertiveness, self management and negotiation skills to navigate the personal and professional environment successfully
CO4	Develop conceptual clarity of the term ' Contributor Personality' and develop understanding of the traits of a contributor
CO5	Develop the qualities of a contributor such as trust, strong work ethic, responsibility and accountability at workplace
CO6	Develop relevant skills required to be a contributor at workplace and achieve success in life

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO3	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO4	-	-	-	-	-	-	-	-	-	-	-	2	-	-
CO5	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO6	-	-	-	-	-	2	-	-	3	-	-	-	-	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

F. Reference Books / Reading

- Contributor Personality Program Workbook (Volume 1,2),
- Contributor Personality Program ActivGuide, Illumine Knowledge Pvt. Ltd.

B. Tech. (Computer Science & Engineering) Programme

SYLLABI (Semester –7)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

Sr No.	Title of the unit	Minimum number of Hours
1.	Introduction to Python Programming & Data Structure	05
2.	Object Oriented Programming	05
3.	Data Handling, Analysis and Visualization	06
4.	Multithreading	04
5.	Network Programming & Web Scrapping	07
6.	Python Database	03
7.	Graphics & GUI Programming	07
8.	Introduction to Web Framework	08

Total Hours (Lab): 30

Total Hours:75

1. Introduction to Python Programming & Data Structure	05 Hours	10%
Basic programming fundamentals, Data variables, Identifies, Operators, Indentation in Python, Indexing, Slicing, BasicString Operations, Using Lists as Stacks, Using Lists as Queues, List Comprehensions, Nested List Comprehensions, Tuples, Sets, Dictionaries, User defined Functions, modules , packages, Python errors, exceptions, raising the exception and debugging concept		
2 Object Oriented Programming	05 Hours	15%
Python Scopes and Namespaces, Class Definition, Class Objects, Instance Objects, Method Objects, Class and Instance Variables, Inheritance, Multiple Inheritance, Private Variables, Polymorphism, Using Properties to Control Attribute Access, Encapsulation and Information Hiding		

3. Data Handling, Analysis and Visualization	06 Hours	15%
Regular expressions with special characters, Regular expressions and raw strings, Extracting matched text from strings, Substituting text with regular expressions, Writing and Reading Binary Data, Writing and Parsing Text Files, Iterating over File Contents Python library cover- RE, Numpy, Pandas & Matplotlib.		
4 Multithreading	04 Hours	10%
Thread, Starting a thread, Threading Module, Synchronizing thread, Multithreaded priority queue		
5. Network Programming & Web Scrapping	07 Hours	10%
Socket, Socket Module Methods for IoT, Client Server Static web scrapping using URLLIB, Beautiful Soup and request library, XML & JSON Parsing, Dynamic web scrapping with selenium and scrapy		
6. Python Database	03 Hours	10%
Database introduction, Connections, Executing queries, Handling Transaction Error, Importing data from Web and CSV		
7. Graphics & GUI Programming	07 Hours	15%
Designing GUI application using Tkinter, PyQt and Kivy		
8. Introduction to Web Framework	08 Hours	15%
Introduction to Django, Model Layer, View Layer, Templet Layer, Forms, Develop Web Service using Flask Framework		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Identify the basic python constructs with a view of using them in problem solving
CO2	Apply control structures and use of data structure, OOPs concepts for real time problem solving
CO3	Analyze utility of strings and functions in modular programming using python and apply exception Handling, file handling and database operations
CO4	To be able to create socket application, GUI application and Web applications in Python
CO5	To be able to create Multithreaded Programming and work in a team

Course Articulation Matrix:

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	1	1	1	1	1	1	1	1	1	2	2	2	3	-
CO2	2	2	2	2	2	2	1	1	2	1	1	1	3	2	-
CO3	3	2	2	1	2	2	1	1	2	2	2	1	2	2	-
CO4	3	2	3	3	3	2	1	1	2	2	2	2	3	3	-
CO5	3	2	2	2	3	2	2	2	2	2	2	2	2	2	-

C. Instructional Methods and Pedagogy:

- At the start, of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Faculty would use coached problem-solving method as it is the class format in which faculty provide a structured, guided context for students working collaboratively to solve problems.
- Assignments based on topic content will be given to the students at the end of each unit/topic.
- Surprise tests/Quizzes will be conducted.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Minimum 30 programs of Python in terms of experiments are suggested in the laboratory.

D. Recommended Study Material:

❖ Text Books:

1. Introducing Python by Lubanovic Bill, O ReILLY
2. Beginning Python: Using Python 2.6 and Python 3.1 by James Payne, Wrox Publication
3. Core Python Programming- Second Edition by Wesley J. Chun

❖ Reference Books:

1. Python GUI Programming- A Complete Reference Guide by Alan D. Moore
2. Django: Web Development with Python by Samuel Dauzon

CS442: DATA SCIENCE AND ANALYTICS

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	4	7	5
Marks	100	100	200	

Pre-requisite courses:

- Data Structures & Algorithm Design
- Database Management System
- Design & Analysis of Algorithms
- Computer Programming
- Engineering Mathematics

A. Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	INTRODUCTION TO DATA SCIENCE	04
2.	STATISTICAL INFERENCE	05
3.	DATA PRE-PROCESSING AND DATA VISUALIZATION	05
4.	INTRODUCTION TO MAP-REDUCE AND HADOOP ARCHITECTURE	05
5.	HDFS, HIVE AND HIVEQL, HBASE	10
6.	APACHE SPARK	06
7.	NoSQL	03
8.	DATA BASE FOR THE MODERN WEB	07
	Total hours (Theory):	45
	Total hours (Lab):	60
	Total hours:	105

B. Detailed Syllabus:

1. INTRODUCTION TO DATA SCIENCE 04 Hours 10%

Introduction of data science and data analytics, defining data science by its key components, Big Data and its importance, Four Vs, Drivers for Big data, Big data applications,

Exploring Data Science in Business, Applications in real-world

- | | | |
|---|-----------------|-------------|
| 2. STATISTICAL INFERENCE | 05 Hours | 08% |
| Event Space, Random Variables and Probability Distributions | | |
| 3. DATA PRE-PROCESSING AND DATA VISUALIZATION | 05 Hours | 15% |
| Dataset, Types of Dataset, Importance of Pre-processing the Data, Data Cleaning, Data Integration and Transformation, Data Reduction, Data Discretization and Concept Hierarchy Generation, Data visualization techniques | | |
| 4. INTRODUCTION TO MAP-REDUCE AND HADOOP ARCHITECTURE | 05 Hours | 12 % |
| Big Data – Apache Hadoop & Hadoop EcoSystem, Moving Data in and out of Hadoop – Understanding inputs and outputs of MapReduce, Data Serialization. | | |
| 5. HDFS, HIVE AND HIVEQL, HBASE | 10 Hours | 20 % |
| HDFS-Overview, Installation and Shell, Java API; Hive Architecture and Installation, Comparison with Traditional Database, HiveQL Querying Data, Sorting And Aggregating, Map Reduce Scripts, Joins & Sub queries, HBase concepts, Advanced Usage, Schema Design, Advance Indexing, PIG, Zookeeper , how it helps in monitoring a cluster, HBase uses Zookeeper and how to Build Applications with Zookeeper. | | |
| 6. Apache SPARK | 06 Hours | 15% |
| Introduction to Data Analysis with Spark, Downloading Spark and Getting Started, Programming with RDDs, Machine Learning with MLlib. | | |
| 7. NoSQL | 03 Hours | 08 % |
| What is it? Where It is Used Types of NoSQL databases, Why NoSQL? Advantages of NoSQL, Use of NoSQL in Industry, SQL vs NoSQL, NewSQL | | |

8. Data Base for the Modern Web**07 Hours 12 %**

Introduction to MongoDB key features, Core Server tools, MongoDB through the JavaScript's Shell, Creating and Querying through Indexes, Document-Oriented, principles of schema design, Constructing queries on Databases, collections and Documents, MongoDB Query Language.

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Use an ethically responsible approach to evaluate and interpret data
CO2	Demonstrate expertise in statistical data processing
CO3	Use of various algorithms as well as mathematical and statistical models and optimization concepts to formulate and the use analyse data appropriately
CO4	Develop the ability to build and evaluate data-based models.
CO5	To learn difference between conventional SQL query language and NoSQL and MongoDB basic concepts
CO6	Utilizing data science principles and approaches to solve real-life situational problems and effectively communicate them.

Course Articulation Matrix:

	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11	PO12
CO1	3	2	1	2	-	-	-	-	-	-	-	-
CO2	1	3	1	-	-	-	-	-	-	-	-	-
CO3	-	2	3	2	-	-	-	-	-	-	-	-
CO4	-	-	3		-	-	-	-	-	-	-	-
CO5	-	2	2	3	-	-	-	-	-	-	-	-
CO6	-	-	-	-	-	-	-	1	-	-	-	3

Recommended Study Material:**❖ Text book:**

1. Data Science and Big Data Analytics: Discovering, Analyzing, Visualizing and Presenting Data, by EMC Education Services, Wiley, 2015.

2. Professional Hadoop Solutions By Boris lublinsky, Kevin t. Smith, Alexey Yakubovich, Wiley, ISBN: 9788126551071, 2015.
3. Understanding Big data By Chris Eaton,Dirkderooset al. , McGraw Hill,2012.
4. BIG Data and Analytics ,Sima Acharya, Subhashini Chhellappan, Willey
5. MongoDB in Action, Kyle Banker,PiterBakkum , Shaun Verch, Dream tech Press
6. HADOOP: The definitive Guide By Tom White, 4th Edition O Reilly 2012.
7. Big Data Analyticswith R and Haoop By VigneshPrajapati, Packet Publishing 2013.
8. Learning Spark: Lightning-Fast Big Data Analysis Paperback by Holden Karau, Apress

❖ **Reference book:**

1. Big Data Analytics with Spark ByGuller, Mohammed,Apress
2. Analytics in a Big Data World: The Essential Guide to Data Science and Its Applications By Bart Baesens, Wiley Publication
3. Hadoop in Practice by Alex Holmes, Manning Publication

❖ **Web material:**

1. <http://www.bigdatauniversity.com/>
2. <https://sparkhub.databricks.com/resources/>

CE442: Design of Language Processors

Credit and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	4	2	-	6	5
Marks	100	50	-	150	

A. Outline of the Course:

Sr No.	Title of the unit	Minimum number of hours
1.	Overview of Language Processors & Translators	10
2.	Introduction To Compilers	20
3.	Symbol-Table Management	05
4.	Static & Dynamic Memory Allocation & Memory management	06
5.	Semantic Analysis & Intermediate Code Generation	05
6.	Code Optimization	07
7.	Code Generation	07

Total hours (Theory): 60.

Total hours (Lab): 30.

Total hours: 90.

B. Detailed Syllabus:

1. Overview of Language Processors & Translators 10 17%

Language processing activities, fundamental of language processing, Operating System, Interpreter vs compiler

Pre-processor & Macro Processors – Subroutine, Macro definition and call, Macro expression, nested macro call, Advanced macro facilities, design of macro pre-processor

Loaders - Compile & go, Absolute, Bootstrap, Relocating, Linking loader

Linkers - Relocation of Linking Concept, Design of Linker, Linker for MS

	DOS, Linking for overlays, Linkage editor Assemblers - Elements of Assembly Language Programming, Assembly Scheme, Single pass Assembler, Two pass assembler, Data structure of Assemblers		
2.	Introduction To Compilers Pass and Phases of Compiler, grouping of phases, Compiler Contraction tools Lexical analyser - Roles of lexical analyser, input buffering, tokens, Regular Expression, finite Automata Syntax analyser - Context free grammar, Ambiguous grammar, Top-down parsing, Bottom-up parsing, LEX, YACC	20	35%
3.	Symbol-Table Management Data structures to implement symbol table, Symbol Attributes, Symbol-Table entries, Local Symbol Table management, Global Symbol Table Structure, Storage bindings and Symbolic Registers	05	08%
4	Static & Dynamic Memory Allocation & Memory management Data descriptors - Static and Dynamic storage allocation – Storage allocation and access in block structured programming languages – Array allocation and access- Compilation of expressions – Handling operator priorities – Intermediate code forms for expressions – code generator., Register Usage, The run-time stack, Parameter passing disciplines, Code sharing and Position-Independent code	06	10%
5.	Semantic Analysis & Intermediate Code Generation Intermediate Languages, Declarations, Assignment Statements, Boolean Expressions, Case Statements, Back patching, Procedure Calls	05	08%
6.	Code Optimization The Principal Sources of Optimization, Optimization of Basic Blocks, Loops in Flow Graphs, Introduction to Global Data-Flow Analysis, Iterative Solution of Data-Flow Equations, Code-Improving Transformations, Dealing with Aliases, Data-Flow Analysis of Structured Flow Graphs, Efficient Data-Flow Algorithms, A Tool for Data-Flow Analysis, Estimation of Types, Symbolic Debugging of Optimized Code	07	11%
7.	Code Generation Issues in the Design of a Code Generator, The Target Machine, Run-Time Storage Management, Basic Blocks and Flow Graphs, Next-Use Information, A Simple Code Generator, Register Allocation and Assignment, The DAG Representation of Basic Blocks, Peephole Optimization, Generating Code from DAGs, Dynamic Programming Code-Generation Algorithm, Code-Generator Generators	07	11%

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand design and processing of different language processor, loaders and linkers
CO2	Design top-down and bottom-up parsers
CO3	Identify different memory management schemes of language processors
CO4	Develop semantic analysis scheme to generate intermediate code
CO5	Apply different code optimization techniques
CO6	Develop algorithms to generate code for a target machine

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	1	-	-	3	-	-	-	-	-	-	-	-	-	-
CO3	2	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	2	2	2	-	2	-	-	-	-	-	-	-	-	-
CO5	2	-	-	-	-	-	-	-	-	-	-	-	-	-
CO6	2	-	-	1	-	-	-	-	-	-	-	-	-	-

C. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries 5 Marks weightage.
- Two internal exams will be conducted and average of the same will be converted to equivalent of 15 Marks as a part of internal theory evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval. It carries a weightage of 5 Marks as a part of internal theory evaluation.
- Surprise tests/Quizzes/Seminar will be conducted which carries 5 Marks as a part of internal theory evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.

❖ Text Books:

- 1 Alfred Aho, Ravi Sethi, Jeffrey D Ullman, “Compilers Principles, Techniques and-Tools”, Pearson Education Asia.
- 2 M. Dhamdhare, “System Programming and Operating Systems”, Tata McGraw-Hill.
- 3 Steven S. Muchnick. Advanced Compiler Design and Implementation.

❖ Reference Books:

- 1 Allen I. Holub “Compiler Design in C”, Prentice Hall of India.
- 2 C. N. Fischer and R. J. LeBlanc, “Crafting a compiler with C”, Benjamin Cummings.
- 3 J.P. Bennet, “Introduction to Compiler Techniques”, Second Edition, Tata McGraw-Hill
- 4 HenkAlblas and Albert Nymeyer, “Practice and Principles of Compiler Building withC”, PHI.
- 5 Kenneth C. Loudon, “Compiler Construction: Principles and Practice”, Thompson Learning.
- 6 Compiler Construction by Kenneth. C. Loudon, Vikas Pub.

❖ Web Links:

1. <http://www.compilerconnection.com>
2. <http://dinosaur.compilertools.net>
3. <http://pltplp.net/lex-yacc>

CE443: CLOUD COMPUTING

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr No.	Title of the unit	Minimum number of Hours
1.	Cluster Computing, Grid Computing Systems and Resource Management at Glance	08
2.	Fundamental of Virtualization	06
3.	Fundamental Concepts and Models	06
4.	Cloud-Enabling Technology	05
5.	Fundamental Cloud Architectures	07
6.	Advanced Cloud Architectures	08
7.	Implementation of Cloud	05

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

- 1 Cluster Computing, Grid Computing Systems and Resource Management at Glance** **08 Hrs 18%**
Introduction, Eras of Computing, Scalable Parallel Computer Architectures, Towards Low Cost Parallel Computing and Motivations, A Cluster Computer and its Architecture, Clusters Classification, Commodity Components for Clusters, Grid Architecture and Service Modelling, Grid Projects and Grid Systems Built, Grid Resource Management and Brokering, Software and Middleware for Grid Computing, Grid Application Trends
- 2. Fundamental of Virtualization** **06 Hrs 12%**
Type of Virtualization, Virtualization Technologies, virtualize your Environment, Managing Virtualization Environment, Storage Virtualization
- 3. Fundamental Concepts and Models** **06 Hrs 15%**
Roles and Boundaries, Cloud Characteristics, Cloud Delivery Models, Cloud Deployment Models

- | | | | |
|----|--|---------------|------------|
| 4. | Cloud-Enabling Technology
Broadband Networks and Internet Architecture, Data Center Technology, Virtualization Technology, Web Technology, Multitenant Technology, Service Technology | 05 Hrs | 14% |
| 5. | Fundamental Cloud Architectures
Workload Distribution Architecture, Resource Pooling Architecture, Dynamic Scalability Architecture, Elastic Resource Capacity Architecture, Service Load Balancing Architecture, Cloud Bursting Architecture, Elastic Disk Provisioning Architecture, Redundant Storage Architecture | 07 Hrs | 16% |
| 6. | Advanced Cloud Architectures
Hypervisor Clustering Architecture, Load Balanced Virtual Server Instances Architecture, Non-Disruptive Service Relocation Architecture, Zero Downtime Architecture, Cloud Balancing Architecture, Resource Reservation Architecture, Dynamic Failure Detection and Recovery Architecture | 08 Hrs | 17% |
| 7. | Implementation of Cloud
Study of Cloud computing Systems like Amazon EC2 and S3, Google App Engine, and Microsoft Azure, Build Private/Hybrid Cloud using open source tools, Deployment of Web Services from Inside and Outside a Cloud Architecture. MapReduce and its extensions to Cloud Computing, HDFS, and GFS. | 05 Hrs | 08% |

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Assess and examine advantages and disadvantages of virtualization technology.
CO2	Compose services in a distributed computing environment to achieve tasks relevant to a knowledge-based business or public service
CO3	Evaluate a set of business requirements to determine suitability for a cloud computing delivery model.
CO4	Explore the various cloud computing architectures and paradigms.
CO5	Deployment of cloud and identify security implications in cloud computing.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	2	1	0	1	-	-	-	1	2	-
CO2	3	3	3	1	3	2	1	-	1	-	1	3	2	1
CO3	3	3	2	3	3	1	2	-	2	1	-	3	2	1
CO4	3	2	1	2	2	-	-	2	1	-	-	2	2	2
CO5	3	2	3	1	3	2	1	1	2	1	-	3	1	1

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

C. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Faculty would use coached problem-solving method as it is class format in which faculty provide a structured, guided context for students working collaboratively to solve problems.
- Attendance is compulsory in lectures and laboratory which carries 5% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval. It carries a weight age of 5%.
- Surprise tests/Quizzes will be conducted which carries 5% component of the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Minimum 10 experiments are suggested in the laboratory related to course content.

D. Recommended Study Material:

❖ Text Books:

1. Thomas Erl, Zaigham Mahmood and Ricardo Puttini, “Cloud Computing Concepts, Technology & Architecture”, Prentice Hall
2. Kai Hwang, Geoffrey C.,” Distributed and Cloud Computing”, MorganKaufmann is an imprint of Elsevier
3. Navin Sabharwal, Ravi Shankar “Apache CloudStack Cloud Computing” PACKT Publishing

❖ Reference Books:

1. Ravi Shankar, Navin Sabharwa “Cloud Computing First Steps: Cloud Computing for Beginners” Create Space Independent Publishing Platform
2. Rajkumar Buyya, James Broberg, Andrzej Goscinski “CloudComputing: Principles and Paradigms” Wiley
3. Judith Hurwitz, Robin Bloor “Cloud Computing For Dummies” , forDummies

CS471: MOBILE COMPUTING (ELECTIVE- III)

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

A. Pre-requisite courses:

- Computer Networks
- Modern Networks

B. Outline of the Course:

Sr No.	Title of the unit	Minimum number of Hours
1.	Wireless Communication Fundamentals	03
2.	Telecommunication Systems	12
3.	Wireless LAN	18
4.	Mobile Network Layer	12
5.	Mobile Transport Layer	12
6.	Advanced Mobile Technologies	03

Total Hours (Theory): 60

Total Hours (Lab):

30 Total Hours: 90

C. Detailed Syllabus:

1.	Wireless Communication Fundamentals	03 Hours	05 %
1.1	Applications		
1.2	A short history of wireless communication		
1.3	A market for mobile communications		
1.4	Some open research topics		
1.5	A simplified referencemodel		
2.	Telecommunication Systems	12 Hours	21 %
2.1	GSM: Mobile services, System architecture, Radio interface, Protocols, Localization and Calling, Handover, Security, New data services		
2.2	DECT: System architecture, Protocol architecture		
2.3	TETRA, UMTS and IMT-2000		

3. Wireless LAN	18 Hours	28 %
31 Infrared vs radio transmission		
32 Infrastructure and ad-hoc network		
33 IEEE 802.11: System architecture, Protocol architecture, Physical layer, Medium access control layer, MAC management, 802.11a, 802.11b, Newer developments		
34 HiperLAN: HiperLAN 1, HiperLAN2		
35 Bluetooth: Architecture, Radio layer, Baseband layer, Link Manager Protocol, L2CAP, Security, SDP, Profiles, IEEE 802.15		
4. Mobile Network Layer	12 Hours	21 %
41 Mobile IP		
42 Dynamic Host Configuration Protocol		
43 Mobile ad-hoc networks: Routing, Destination sequence distance vector, Dynamic source routing		
5. Mobile Transport Layer	12 Hours	21 %
51 Traditional TCP: Congestion control, Slow start, Fast retransmit/fast recovery, Implications of mobility		
52 Classical TCP improvements: Indirect TCP, Snooping TCP, Mobile TCP, Fast retransmit/fast recovery, Transmission/time-out freezing, Selective retransmission, Transaction-oriented TCP		
53 TCP over 2.5/3G wireless networks		
6. Advanced Mobile Technologies	03 Hours	04 %
61 Introduction of 4G, 5G and other advanced mobile technologies		
62 The architecture of future networks		

D. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries 5 Marks weightage.
- Two internal exams will be conducted and average of the same will be converted to equivalent of 15 Marks as a part of internal theory evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval. It carries a weightage of 5 marks as a part of internal theory evaluation.
- Surprise tests/Quizzes/Seminar will be conducted which carries 5 Marks as a part of internal theory evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

- Experiments/Tutorials related to course content will be carried out in the laboratory.

E. Course Outcome (COs):

CO1	Demonstrate the principles and theories of mobile computing technology
CO2	Characterize the infrastructure and technology of mobile computing technologies
CO3	List applications in various domains offered to the public, employees and businesses through mobile computing technology
CO4	Describes the current technologies to build potential future of mobile computing technologies and applications
CO5	Identify the potential future of mobile computing technologies and applications

F. Course Articulation Matrix:

	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11	PO12
CO1	3	1	-	-	-	-	-	-	-	-	-	-
CO2	1	2	1	-	-	-	-	-	-	-	-	-
CO3	1	2	2	-	2	-	-	-	-	-	1	-
CO4	-	3	1	1	-	-	1	-	-	-	-	-
CO5	-	3	1	1	-	-	1	-	-	-	-	-

G. Recommended Study Material:

❖ Text Books:

1. Jochen Schiller, "Mobile Communications", PHI/Pearson Education, Second Edition, 2003.
2. "Mobile Computing: Technology, Applications and Service Creation" by Asoke K Talukder and Roopa R Yavagal, TMH, ISBN: 0-07-058807-4

❖ Reference Materials:

1. William Stallings, "Wireless Communications and Networks", PHI/Pearson Education, 2002.
2. Kaveh Pahlavan, Prasanth Krishnamoorthy, "Principles of Wireless Networks", PHI/Pearson Education, 2003.
3. Uwe Hansmann, Lothar Merk, Martin S. Nicklons and Thomas Stober, "Principles of Mobile Computing", Springer, New York, 2003.
4. Hazysztof Wesolowski, "Mobile Communication Systems", John Wiley and Sons Ltd, 2002
5. Research papers from IEEE, Springer etc.

❖ Web Materials:

1. www.ietf.org – For drafts
2. www.ieee.org – For standards and technical research papers

CS472: CYBER SECURITY AND CYBER LAW (ELECTIVE- III)

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

A. Outline of the Course:

Sr No.	Title of the unit	Minimum number of Hours
1	Introduction to Cyber Security	04
2	Cyber Security Vulnerabilities and Cyber Security Safeguards	09
3	Securing Web Application, Services and Servers	05
4	Intrusion Detection and Prevention	12
5	Cryptography and Network Security	09
6	Introduction to Cyberspace and the Law	03
7	Cyber Forensic	03

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

- 1 Introduction to Cyber Security 04 hours 10%**
 - 1.1 Overview of Cyber Security
 - 1.2 Internet Governance – Challenges and Constraints
 - 1.3 Cyber Threats:- Cyber Warfare-Cyber Crime-Cyber terrorism-Cyber Espionage
 - 1.4 Need for a Comprehensive Cyber Security Policy, Need for a Nodal Authority, Need for an International convention on Cyberspace.
- 2 Cyber Security Vulnerabilities and Cyber Security Safeguards 09 hours 20%**
 - 2.1 Cyber Security Vulnerabilities-Overview
 - 2.2 Vulnerabilities in software, System administration, Complex Network Architectures, Open Access to Organizational Data, Weak Authentication, Unprotected Broadband communications, Poor Cyber Security Awareness.
 - 2.3 Cyber Security Safeguards- Overview
 - 2.4 Access control, Audit, Authentication, Biometrics, Cryptography, Deception, Denial of Service Filters
 - 2.5 Ethical Hacking, Firewalls, Intrusion Detection Systems, Response, Scanning, Security policy, Threat Management.

3	Securing Web Application, Services and Servers	05 hours	10%
3.1	Introduction, Basic security for HTTP Applications and Services, Basic Security for SOAP Services		
3.2	Identity Management and Web Services, Authorization Patterns		
3.3	OWASP Top 10, Security Considerations, Challenges		
4	Intrusion Detection and Prevention	12 hours	25%
4.1	Introduction of Intrusion Detection and Prevention, Physical Theft, Abuse of Privileges, Unauthorized Access by Outsider, Malware infection		
4.2	Intrusion detection and Prevention Techniques, Anti-Malware software		
4.3	Network based Intrusion detection systems, Network based Intrusion Prevention Systems		
4.4	Host based Intrusion prevention Systems, Security Information Management		
4.5	Network Session Analysis, System Integrity Validation		
5	Cryptography and Network Security	09 hours	25%
5.1	Introduction to Cryptography		
5.2	Symmetric key Cryptography, Asymmetric key Cryptography,		
5.3	Message Authentication, Digital Signatures, Applications of Cryptography.		
5.4	Overview of Firewalls- Types of Firewalls, User Management		
5.5	VPN Security Protocols: - security at the Application Layer- PGP and S/MIME		
5.6	Security at Transport Layer- SSL and TLS, Security at Network Layer-IPSec		
6	Introduction to Cyberspace and the Law	03 hours	05%
6.1	Introduction, Cyber Security Regulations		
6.2	Roles of International Law, the state and Private Sector in Cyberspace, Cyber Security Standards		
6.3	The INDIAN Cyberspace, National Cyber Security Policy 2013.		
7	Cyber Forensics	03 hours	05%
7.1	Introduction to Cyber Forensics, Handling Preliminary Investigations, Controlling an Investigation		
7.2	Conducting disk-based analysis, Investigating Information-hiding		
7.3	Scrutinizing E-mail, Validating E-mail header information, Tracing Internet access, Tracing memory in real-time.		

Course Outcome(Co):

At the end of the course, the students will be able to

CO1	After learning the course, students should able to learn about cyber-attack, type of cybercrimes
CO2	Learn cyber security vulnerabilities and security safeguard, prevent the web using different web application tools.
CO3	Learn about how to protect the network using different network tools and Intrusion detection and prevention techniques, Learn basis of Cryptography and network security
CO4	Understand about cyber laws
CO5	Learn about cyber Forensics and investigation of information hiding.
CO6	Learn about how to protect them self and ultimately society from cyber- attacks.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	-	3	-	-	-	-	-	-	2	-
CO2	-	3	2	2	3	2	2	-	-	-	-	-
CO3	3	-	3	2	3	2	2	2	-	-	-	-
CO4	-	-	-	-	-	2	-	3	-	2	-	2
CO5	-	-	3	3	3	-	-	3	-	-	2	-
CO6	2	-	-	-	-	3	-	-	3	-	-	2

C. Instructional Methods and Pedagogy:

- At the start, of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Faculty would use coached problem-solving method as it is the class format in which faculty provide a structured, guided context for students working collaboratively to solve problems.
- Assignments based on topic content will be given to the students at the end of each unit/topic.
- Surprise tests/Quizzes will be conducted.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures..

D. Recommended Study Material:

◆ **Text Books:**

1. Cybersecurity: The Beginner's Guide Dr. Erdal Ozkaya

◆ **Reference Books:**

3. Digital Privacy and Security Using Windows: A Practical Guide ByNihad Hassan, Rami Hijazi, Apress
4. Cyber Security Understanding Cyber Crimes, Computer ForensicsandLegal Perspectives by Nina Godbole and Sunit Belpure, Publication Wiley

CS473: Block Chain Technology & Applications

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

A. Outline of the Course:

Sr No.	Title of the unit	Minimum number of Hours
1	Backbones of Blockchain: Cryptography and Hashing	07
2	An Introduction to Blockchain Technology	06
3	Block-Chain Consensus Mechanisms	12
4	Introduction to Cryptocurrency: BitCoin	12
5	Blockchain hyperledger	05
6	Solidity Programming and Decentralized Application	08
7	Blockchain in Real World	10

Total Hours (Theory): 60

Total Hours (Lab): 30

Total Hours: 80

B. Detailed Syllabus:

- 1 Backbones of Blockchain: Cryptography and Hashing** **07 hours** **12%**
Symmetric key cryptography, Public key cryptography, Digital Signature, Cryptographically Secured Hash Functions, Cryptographically Secured Chain of Blocks, Merkle Trees.
- 2 An Introduction to Blockchain Technology** **06 hours** **10%**
Evaluation of Blockchain Technology, Distributed Systems, The History of Blockchain and Bitcoin, Types of Block-Chain
- 3 Block-Chain Consensus Mechanisms** **12 hours** **20%**
Practical Byzantine fault tolerance algorithm, Proof of Work, Proof of Stake, Proof of Authority, Proof of Elapsed time
- 4 Introduction to Cryptocurrency: BitCoin** **12 hours** **20%**
Digital keys and addresses, Transactions, Mining, The Bitcoin network, Wallets, Bitcoin payments

- 5 **Blockchain hyperledger** **5 hours** **8%**
 Fabric architecture, implementation, networking, fabric transactions, demonstration, smart contracts
- 6 **Solidity Programming and Decentralized Application** **08 hours** **13%**
 Using the Remix IDE, Data Types and Functions, Creating Inline Assembly Functions, Mappings, Modifiers, Structs, and More, Generating ERC-20 Tokens, Extending Token Security (ERC-223), Deploying the ERC-20 Token Contract, DAPP using Ehtereum, DAPP using Hyperledger.
- 7 **Blockchain in Real World** **10 hours** **17%**
 Blockchain applications, eGovernance, smart cities, smart industries, use cases, trends on blockchains, serverless blocks, scalability issues, blockchain on clouds, Blockchain in IOT.

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand the Basic Cryptography behind the Blockchain Technology and Bitcoin.
CO2	Define the structure of a Blockchain and classify why and when it is better than a simple distributed database.
CO3	Analyse the consensus mechanisms in a Blockchain Technology and critically assess its applicability in Blockchain based application.
CO4	Analyse to what extent smart and self-executing contracts can benefit automation, governance and transparent environment.
CO5	Design decentralized distributed application and measure the performance of Blockchain against centralized system.
CO6	Attain awareness of the new challenges that exist in monetizing businesses around Blockchain.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	-	-	1	-	-	2	1	-	-	-	2	-
CO2	-	2	-	-	-	2	1	1	1	-	2	-	1	2
CO3	1	3	2	1	-	2	-	-	2	-	1	2	3	-
CO4	1	1	2	2	3	2	2	2	2	-	3	2	3	1

CO5	1	3	3	2	3	2	2	2	3	2	3	2	3	2
CO6	1	3	1	3	-	2	2	-	3	1	3	2	3	2

C. Instructional Methods and Pedagogy:

- At the start, of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Faculty would use coached problem-solving method as it is the class format in which faculty provide a structured, guided context for students working collaboratively to solve problems.
- Assignments based on topic content will be given to the students at the end of each unit/topic.
- Surprise tests/Quizzes/Seminars will be conducted.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

D. Recommended Study Material:

❖ Text Books:

1. Imran Bashir , Mastering Blockchain ,Packt Second Edition, 2018.
2. Andreas M. Antonopoulos, Mastering Bitcoin, O'Reilly, Second Edition

❖ Reference Books:

1. Samanyu, Blockchain Developer's Guide, Packt, 2018.
2. Mark Watney, Blockchain for Beginners: The Complete Step by Step Guideto Understanding Blockchain Technology, July, 2017.
3. Don Tapscott, Blockchain Revolution: How the Technology Behind Bitcoin Is Changing Money, Business, and the World, Hardcover, May 2016.
4. Xun (Brian) Wu, Hyperledger Cookbook, Packt.
5. Mayukh Mukhopadhyay, Ethereum Smart Contract Development, Packt,2018.

❖ Web Materials:

<https://medium.com/topic/blockchain>

CS445: SOFTWARE GROUP PROJECT - IV

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	0	2	2	2
Marks	0	50	50	

A. Outline of the Course:

- Student at the beginning of a semester may be advised by his/her supervisor (s) for recommended courses.
- Students will work together in a team (at most **three**) with any programming language.
- Students are required to get approval of project definition from the department.
- After approval of project definition students are required to report their project work on weekly basis to the respective internal guide.
- Project will be evaluated at least once per week in laboratory Hours during the semester and final submission will be taken at the end of the semester as a part of continuous evaluation.
- Project work should include whole SDLC of development of software / hardware system as a solution of particular problem by applying principles of Software Engineering.
- Students have to submit project with following listed documents at the time of final submission.
 - a. Project Synopsis
 - b. Software Requirement Specification
 - c. SPMP
 - d. Final Project Report
 - e. Project Setup file with Source code
 - f. Project Presentation (PPT)
- A student has to produce some useful outcome by conducting experiments or project work.

Total Hours (Theory): 00

Total Hours (Lab):

30 Total Hours: 30

B. Instructional Method and Pedagogy:

- Students will select related topic based on subjects they learnt and other literatures like books, periodicals, journals and various internet resources.
- Students can select the topic based on the research areas of available supervisor/guide.
- Each student has to prepare a synopsis, SRS, SPMP, Design documents and Final Report in prescribed format only. The report typed on A4 sized sheets and bound should be submitted after approval by the supervisor/guide and endorsement of the Head of Department.
- Performance of students at the Project will be assessed by the department faculty. The supervisor/guide based on the quality of work and preparation and understanding of the candidate shall do an assessment of the project.

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Identify problems present in society by surveying variety of domains and convert in project definition.
CO2	Explore new ideas and techniques to solve it. Create, select and apply appropriate techniques, resources, modern engineering and IT tools to solve problem.
CO3	Correlate knowledge of different subjects and apply theoretical knowledge to implement project for identified problem.
CO4	Apply ethical principles and commit to responsibilities and norms of the project.
CO5	Write technical report and deliver presentation by applying different visualization tools and evaluation metrics.
CO6	Apply engineering and management principles to achieve project goal.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	3	-	2	-	-	-	-	-	-	-	-	-	-
CO2	-	-	3	-	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-	-	3	-
CO4	-	-	-	-	-	-	-	3	2	-	-	-	-	-
CO5	-	-	1	-	3	2	-	-	-	3	-	-	-	-
CO6	3	2	2	1	2	1	-	-	2	3	-	1	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

C. Recommended Study Material:

1. Reading Materials, web materials, Project reports with full citations
2. Books, magazines & Journals of related topics
3. Various software tools and programming languages compiler related to topic

❖ Web Materials:

1. www.ieeexplore.ieee.org
2. www.sciencedirect.com
3. www.elsevier.com
4. <http://spie.org/x576.xml>

CS446: SUMMER INTERNSHIP-II

Credit and Hours:

Teaching Scheme	Project	Practical	Tutorial	Total	Credit
Hours/week	90	-	-	90	3
Marks	150	-	-	150	

A. Outline of the Course:

1.	Instructional Method and Pedagogy
	• Summer internship shall be at least 90 hours during the summer vacation only. • Department/Institute will help students to find an appropriate company/industry/organization for the summer internship. • The student must fill up and get approved a Summer Internship Acceptance form by the company and provide it to the Coordinator of the department within the specified deadline. • Students shall commence the internship after the approval of the department Coordinator. Summer internships in research centers is also allowed. • During the entire period of internship, the student shall obey the rules and regulations of the company/industry/organization and those of the University. • Due to inevitable reasons, if the student will not able to attend the internship for few days with the permission of the supervisor, the department Coordinator should be informed via e-mail and these days should be

	compensated later. • The student shall submit two documents to the Coordinator for the evaluation of the summer internship: • Summer Internship Report • Summer Internship Assessment Form • Upon the completion of summer internship, a hard copy of “Summer Internship Report” must be submitted through the presentation to the Coordinator by the first day of the new term. • The report must outline the experience and observations gained through practical internship, in accordance with the required content and the format described in this guideline. Each report will be evaluated by a faculty member of the department on a satisfactory/unsatisfactory basis at the beginning of the semester. • If the evaluation of the report is unsatisfactory, it shall be returned to the student for revision and/or rewriting. If the revised report is still unsatisfactory the student shall be requested to repeat the summer internship.
2.	Format of Summer Internship Report
	The report shall comply with the summer internship program principles. Main headings are to be centered and written in capital boldface letters. Sub-titles shall be written in small letters and boldface. The typeface shall be Times New Roman font with 12pt. All the margins shall be 2.5cm. The report shall be submitted in printed form and filed. An electronic copy of the report shall be recorded in a CD and enclosed in the report. Each report shall be bound in a simple wire vinyl file and contain the following sections: • Cover Page • Page of Approval and Grading • Abstract page: An abstract gives the essence of the report (usually less than one page). Abstract is written after the report is completed. It must contain the purpose and scope of internship, the actual work done in the plant, and conclusions arrived at. • TABLE OF CONTENTS (with the corresponding page numbers)

- LIST OF FIGURES AND TABLES (with the corresponding page numbers)
- DESCRIPTION OF THE COMPANY/INDUSTRY/ORGANISATION:
Summarize the work type, administrative structure, number of employees (how many engineers, under which division, etc.), etc. Provide information regarding
 - Location and spread of the company
 - Number of employees, engineers, technicians, administrators in the company
 - Divisions of the company
 - Your group and division
 - Administrative tree (if available)
 - Main functions of the company
 - Customer profile and market share
- INTRODUCTION: In this section, give the purpose of the summer internship, reasons for choosing the location and company, and general information regarding the nature of work you carried out.
- PROBLEM STATEMENT: What is the problem you are solving, and what are the reasons and causes of this problem.
- SOLUTION: In this section, describe what you did and what you observed during the summer internship. It is very important that majority of what you write should be based on what you did and observed that truly belongs to the company/industry/organization.
- CONCLUSIONS: In the last section, summarize the summer internship activities. Present your observations, contributions and intellectual benefits. If this is your second summer internship, compare the first and second summer internships and your preferences.
- REFERENCES: List any source you have used in the document including books, articles and web sites in a consistent format.
- APPENDICES: If you have supplementary material (not appropriate for the main body of the report), you can place them here. These could be schematics, algorithms, drawings, etc. If the document is a datasheet and it

	can be easily accessed from the internet, then you can refer to it with the appropriate internet link and document number. In this manner you don't have to print it and waste tons of paper.
	Total hours (Project) : 90
	Total hours : 90

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Ability to integrate existing and new technical knowledge for industrial application.
CO2	Executing work with team and teammates from other disciplines
CO3	Get practices and experience related to professional and ethical issues in the work environment
CO4	Experience of demonstrating the impact of the internship on their learning and professional development.
CO5	Understanding of lifelong learning processes through critical reflection of internship experiences.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	1	-	-	1	-	-	-	-	-	-	-	-	-
CO2	-	-	-	-	-	-	-	-	3	2	1	-	-	2
CO3	-	-	-	-	-	-	1	3	-	1	-	-	-	1
CO4	-	-	-	-	-		3	1	-	-	-	-	1	-
CO5	-		-	-	-	1	-	-	-	-	-	3	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ Reference book:

1. Books, Magazines & Journals of related topics

❖ Web material:

1. www.ieeexplore.ieee.org
2. www.sciencedirect.com
3. www.elsevier.com
4. <http://spie.org/x576.xml>

❖ Software

1. ASP.NET
2. PYTHON/MATLAB
3. PHP
4. ANDROID/IOS
5. FLUTTER
6. NODE/REACT NATIVE

B. Tech. (Computer Science & Engineering) Programme

SYLLABI (Semester – 8)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CS447: SOFTWARE PROJECT MAJOR

Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	0	36	36	20
Marks	0	600(250+350)	600	

A. Outline of the Course:

- Software Project includes course work on a specialized Subject or a Seminar.
- The course work shall be related to the area of his/her project research work.
- Students have to take 3 months training to the other software industry as the project work.
- The major project work provides students an opportunity to do something on their own and under the supervision of internal guide as well as guide from industry.
- Student at the beginning of a semester may be advised by his/her supervisor (s) for recommended courses.
- Project will be evaluated at least thrice during the semester by internal guide of the project and final submission at the end of the semester as a part of continuous evaluation.
- Project work should include whole SDLC of development of software / hardware system as solution of particular problem by applying principles of Software Engineering.
- A student has to produce some useful outcome by conducting experiments or project work.
- Student can learn all aspects & functionality of specialized software from the industry.
- Students have to submit SRS, SPMP, Design documents, Code and Test Cases in form of Project report.

B. Instructional Method and Pedagogy:

- Students can select the topic based on the industry requirement.
- Students can select the topic based on the research areas of available supervisor/guide.
- Each student has to prepare a synopsis, SRS, SPMP, Design documents and Final Report in prescribed format only. The report typed on A4 sized sheets and bound should be submitted after approval by the guide and endorsement of the Head of Department.
- Performance of students at the Project will be assessed by the department faculty. The supervisor/guide based on the quality of work and preparation and understanding of the candidate shall do an assessment of the project.
- Following format for documentation for the project be followed:
 - A. Forwarding Page
 - 1. Title of the project
 - 2. Objectives
 - 3. Definitions of Key Term Approach to Problem solving Limitations, if any
 - 4. Output Generated
 - 5. Details of Hardware Performed
 - 6. Details of Software Tools used
 - 7. Implementation issues (clearly defining the area of application)
 - 8. Miscellaneous
 - 9. Signature of candidate & date

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Identify and justify/analyse the requirements of the projects with enhancement of the required tools and technology by individual and team.
CO2	Solve challenging projects for commercial, societal and environment benefit.
CO3	Apply the engineering knowledge to full fill the requirements of the projects pertaining to any discipline.
CO4	Explain the importance of planning, documentation, punctuality and work ethics.
CO5	Document the work which is carried out in proper format with industry standards.
CO6	Showcase the soft skill.

Course Articulation Matrix:

	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	-	2	3	-	1	-	3	1	1	1	3	1
CO2	2	-	3	-	-	3	2	-	-	-	-	-	2	2
CO3	3	2	1	1	1	-	-	-	1	-	2	1	3	1
CO4	3	1	2	-	-	-	-	3	-	-	1	2	2	3
CO5	3	-	2	2	-	-	-	-	-	2	-	3	1	1
CO6	1	-	1	1	-	-	-	3	1	3	1	1	1	3

C. Recommended Chapters/sections

1. Microscope Summery
- 2.Details of candidate and supervisor along with certificate of
 - original work;
 - Assistance, if any;
 - Credits;
3. Aims and Objectives
4. Approaches to Project and Time Frame
5. Project Design Description with appendices to cover
 - Flow charts/Data Flow Diagram – Macro/Micro Level
 - Source code, If any
 - Hardware platform
 - Software Tools
 - Security Measures
 - Quality Assurance
 - Audit ability
6. Test Date and Result

D. Recommended Study Material:

1. Reading Materials, web materials, Project reports with full citations
2. Books, magazines & Journals of related topics
3. Various software tools and programming languages compiler related to topic

❖ Web Materials:

1. www.ieeexplore.ieee.org
2. www.sciencedirect.com
3. www.elsevier.com
4. <http://spie.org/x576.xml>